

Uni-RAT: A Unified DECT-2020 NR and Wi-Fi Framework for 6G Multi-RAT Networks

Yongchao Dang, Ngo Quang, Ruttik Kalle, Jose Costa Requena, and Riku Jäntti

Abstract—Future 6G networks aim to deliver seamless, secure, and low-latency connectivity across heterogeneous radio access technologies (RATs) such as 5G New Radio (NR), Wi-Fi, and DECT-2020 NR to support emerging applications including extended reality (XR) and the Industrial Internet of Things (IIoT). However, current Multi-RATs systems are operated and managed in isolation, which hinders users and devices from seamlessly leveraging heterogeneous connectivity in a unified and efficient manner. To address this fragmentation, 3GPP has introduced interworking functions such as the N3IWF for untrusted and the TNGF for trusted non-3GPP access, enabling IP-session continuity when users move across heterogeneous networks. Nevertheless, these mechanisms rely on mandatory IPsec, which requires each RAT to maintain a separate secure tunnel to the core network and prevents intra-cell device-to-device (D2D) communication, thereby introducing significant encryption latency, increasing data-plane load, and complicating network management. To overcome these limitations, we propose Uni-RAT, a unified Multi-RATs framework in which DECT-2020 NR operates as the Master-RAT, anchoring 3GPP control-plane security via 5G-AKA and NAS, while Wi-Fi functions as a Secondary-RAT for data-plane data delivery. Uni-RAT employs lightweight Layer-3 WireGuard tunnels to secure both D2D and multi-hop communication, significantly reducing the latency and computational overhead of IPsec-based interworking mechanisms used by N3IWF and TNGF. In addition, a customized SMS-over-NAS signaling mechanism is introduced to distribute WireGuard configuration and keying material, enabling unified authentication and efficient Multi-RATs security management. We implement a full prototype using Nordic nRF9161 DECT-2020 NR devices, Raspberry Pi Wi-Fi nodes, and a Cumucore 5G core, with security-critical functions protected inside Arm TrustZone. Experimental results indicate that Uni-RAT supports Multi-RATs communication with high throughput and low packet loss in intra-cell D2D scenarios, maintains bounded latency and jitter under relay-assisted and inter-cell configurations, and achieves stable UE registration latency below 100 ms with up to 100 concurrent devices under background traffic, demonstrating its practical feasibility and scalability and positioning Uni-RAT as an early architectural blueprint for 6G Multi-RATs networks.

Index Terms—Multi-RATs, Wi-Fi, DECT-2020 NR, Device-to-Device (D2D), SMS-over-NAS, 6G.

I. INTRODUCTION

Emerging 6G services, such as extended reality (XR) and the Industrial Internet of Things (IIoT), demand seamless wireless connectivity characterized by extreme data rates, ultra-low latency, and high reliability [1]. However, existing radio access technologies (RATs) cannot seamlessly meet these diverse requirements across heterogeneous deployment environments.

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For instance, Wi-Fi primarily targets indoor environments and achieves low deployment cost by operating in unlicensed spectrum, which, however, makes it susceptible to interference and security vulnerabilities [2]. In contrast, mobile cellular networks such as 5G offer enhanced security, reliability, and quality of service through the use of licensed spectrum and network slicing, but at the expense of higher capital expenditure and dependence on vendor-specific infrastructure [3]. Beyond Wi-Fi and 5G, DECT-2020 NR, recognized by the ITU as a non-cellular technique of the IMT-2020 (5G) family, has emerged as a promising technology for massive machine-type communications (mMTC) and ultra-reliable low-latency communications (URLLC) in industrial and mission-critical applications [4]. According to the ITU, future 6G systems will adopt a Multi-RATs architecture, where cellular networks serve as the primary Big-RAT in coordination with multiple Lite-RATs, consist of non-cellular techniques such as Wi-Fi and DECT-2020 NR, to support diverse use cases while ensuring consistent performance, interoperability, and cost-effective connectivity [5].

The 3rd generation partnership project (3GPP) specifies two interworking functions for non-3GPP access in TS23.502 [6], which extend 5G connectivity beyond licensed cellular access and enable the integration of heterogeneous RATs such as Wi-Fi and LiFi with the 5G Core (5GC). In the untrusted non-3GPP access model, the non-3GPP interworking function (N3IWF) connects the user equipment (UE) to the 5GC via an end-to-end (E2E) IPsec tunnel that secures both NAS signaling and data-plane traffic. In contrast, the trusted non-3GPP access model employs the Trusted non-3GPP gateway function (TNGF), which enables direct NAS signaling between the UE and the access and mobility management function (AMF) while relying on trusted link-layer security mechanisms, such as IEEE 802.1X or MACsec. While both N3IWF and TNGF enable non-3GPP access to the 5GC, they are fundamentally designed to integrate each non-3GPP RAT independently through per-access secure connectivity to the core network. Consequently, security contexts are not shared across RATs, which prevents coordinated cross-RAT control and results in fragmented control-plane operation. Moreover, although these mechanisms ensure E2E integrity and confidentiality, their reliance on IPsec tunneling or link-layer security increases configuration complexity and computational overhead, which limitations constrain deployment flexibility in heterogeneous environments, particularly for time-critical and resource-constrained IIoT scenarios envisioned for future 6G networks.

A promising approach to addressing these limitations is a hierarchical Multi-RATs coordination framework, in which one RAT is designated as a Master-RAT that maintains a persistent and secure connection to the 5GC for authentication and control, while Secondary-RATs operate under a shared security context to provide data-plane connectivity. Related concepts can be observed in commercial systems such as AirDrop and Intel Unison, where Bluetooth low energy (BLE) is used for device discovery and signaling while Wi-Fi delivers high-speed data transfer [7]. However, these D2D solutions operate in isolation and lack integration with a unified 3GPP control and security framework, making them vulnerable to spoofing, relay, and man-in-the-middle attacks [8]–[10]. While existing 3GPP interworking mechanisms such as N3IWF and TNGF could, in principle, connect multiple RATs (e.g., BLE and Wi-Fi) to the 3GPP network, they require independent authentication and management procedures for each RAT, thereby increasing signaling complexity and processing overhead. In contrast, by anchoring security and control at the Master-RAT, a hierarchical architecture enables unified security management, cross-RAT key distribution, and coordinated operation among subordinate Lite-RATs, providing a scalable, low-latency, and secure foundation for Multi-RATs integration in future 6G networks.

Building on our prior work on enhanced wireless direct connectivity and DECT-2020 NR base-station design [7], [11], we present Uni-RAT, a unified Multi-RATs framework that integrates DECT-2020 NR and Wi-Fi into a single control, authentication, and security domain, where DECT-2020 NR acts as the Master-RAT to anchor 3GPP security via 5G-AKA and NAS and distributes the security context to Wi-Fi as a coordinated high-throughput data-plane RAT, thereby mitigating fragmentation in heterogeneous Multi-RATs access architectures. Uni-RAT employs a lightweight, kernel-level Layer-3 WireGuard encryption mechanism with minimal code complexity to efficiently secure both control-plane and data-plane traffic between the 5GC and UEs, as well as D2D communication, significantly reducing latency and computational overhead while preserving strong security. Moreover, a dedicated WireGuard application function (WG-AF) is implemented in the core network to manage and distribute WireGuard configuration and keying material via short message service over the 5G non-access stratum (SMS-over-NAS), enabling rapid and secure data-plane establishment while maintaining bounded latency and jitter in intra-cell D2D, relay-assisted one-hop, and inter-cell multi-hop scenarios. The main contributions of this paper are summarized as follows:

- We propose Uni-RAT, a unified Multi-RATs framework in which DECT-2020 NR acts as a Master-RAT to anchor 3GPP authentication and NAS security, which is shared with coordinated Wi-Fi secondary-RATs for high-throughput data delivery. By decoupling control-plane security from data-plane transmission, Uni-RAT enables Wi-Fi to inherit 3GPP-grade security without redundant authentication, eliminating per-RAT fragmentation and

supporting scalable, secure Multi-RATs operation for 6G networks.

- Uni-RAT replaces the IPsec-based N3IWF and TNGF with lightweight Layer-3 WireGuard tunnels, enabling secure communication between UEs and the 5GC as well as D2D connectivity. To support scalable key management, Uni-RAT introduces a customized SMS-over-NAS signaling mechanism to distribute WireGuard configuration and keying material to devices, enabling unified security control across heterogeneous RATs for both control and data planes while reducing cryptographic overhead and computational load.
- Uni-RAT introduces a hardware-assisted security architecture based on ARM TrustZone at both UEs and base station (BS), enabling secure integration of heterogeneous RATs under a unified 3GPP control framework. Security-critical functions, including eSIM handling, 5G-AKA, NAS key derivation, and WireGuard key management, are executed exclusively within the TrustZone on the DECT-2020 NR Master RAT and are exposed to Wi-Fi Secondary RATs only through controlled service interfaces, enabling secure cross-RAT security-context sharing without exposing control-plane credentials.
- A Uni-RAT prototype was implemented using Nordic nRF9161 development boards, Raspberry Pi platforms, and a Cumucore 5G Core. On the device side, the Nordic platform supports DECT-2020 NR and provides an ARM TrustZone execution environment for security-critical operations, compensating for the lack of hardware security support on Raspberry Pi devices acting as Wi-Fi-based Secondary-RATs. Within the 5GC, a customized SMS-over-NAS procedure and a dedicated WG-AF were implemented for WireGuard configuration and key distribution. Experimental results demonstrate full system interoperability and stable operation with up to 100 concurrent RAT devices, validating Uni-RAT's practicality, scalability, and robustness as a foundation for future 6G heterogeneous access systems.

The remainder of this paper is organized as follows. Section II reviews background topics, including multi-RAT technologies, multi-RAT cooperation and integration, and intelligent multi-RAT management and orchestration. Section III presents the Uni-RAT framework, including the system architecture and control-plane and data-plane signaling. Section IV presents the Uni-RAT architecture and procedures across the UE, BS, and 5GC. Section V evaluates the Uni-RAT implementation and performance using a protocol testbed based on DECT-2020 NR development boards, Raspberry Pi devices, and the Cumucore 5GC, including data-plane performance analysis of WireGuard D2D and multi-hop communication. Finally, Section VI discusses limitations of the current work, outlines future research directions, and concludes the paper with perspectives on multi-RAT in 6G networks.

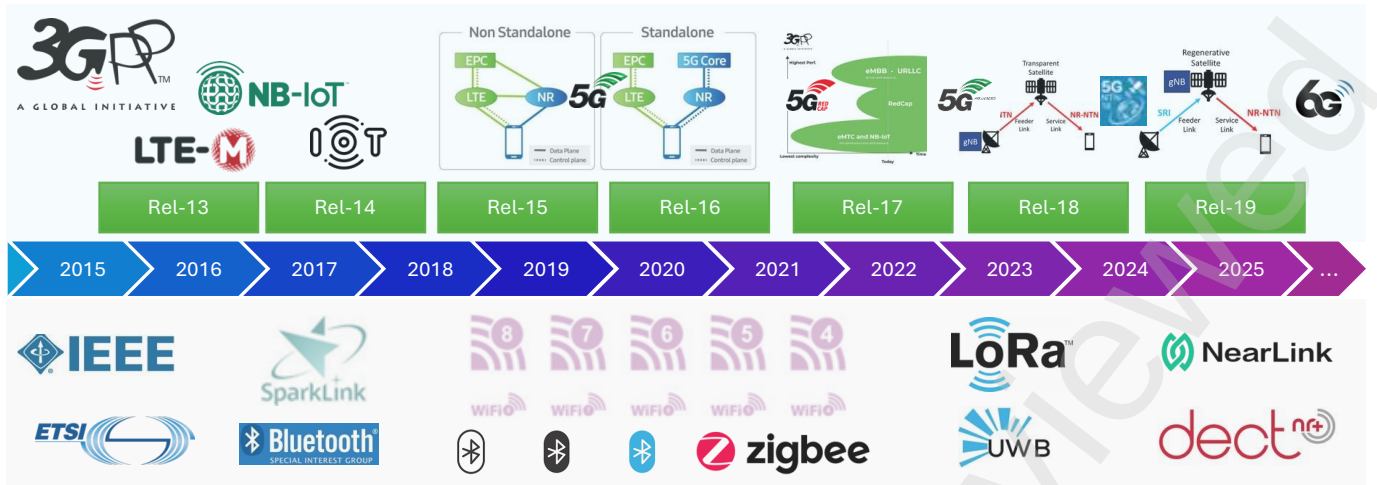


Fig. 1: 3GPP and non-3GPP RATs.

II. RELATED WORK

This section reviews the evolution of multi-RAT technologies across 3GPP and non-3GPP domains, covering advances in cooperation, integration, intelligent management and orchestration, and security.

A. Multi-RATs Technology Evolution

1) *3GPP RATs*: Over the past decade, the rapid growth of wireless data traffic and the increasing demand for higher performance and lower cost have driven 3GPP to evolve its connectivity technologies beyond legacy systems such as 2G and 3G, which can no longer meet the diverse requirements of emerging IoT and machine-type communication applications [12]. From Release 13 (2016) to Release 19 (2025), 3GPP has progressively extended connectivity from human-centric broadband to large-scale machine communications. Release 13 introduced NB-IoT and LTE-M (eMTC) to enable mMTC with low power and cost [13], while Release 14 enhanced mobility, positioning, and multicast capabilities within the Cellular IoT (CIoT) framework. Release 15 (2018) marked the 5G era with NR and the service-based 5GC in Non-Stand-Alone (NSA) mode, later evolving in Release 16 (2020) to fully integrate NB-IoT/LTE-M as native 5G RATs and introduce NR-based V2X via sidelink [14], [15]. Release 17 (2022) extended 5G to Non-Terrestrial Networks (NTN), standardizing NB-IoT-over-NTN and NR-over-NTN for satellite IoT, and introduced Reduced-Capability NR (RedCap) for mid-tier IoT devices [16], [17]. Release 18 (2024), the first 5G-Advanced phase, emphasized energy efficiency, enhanced RedCap (eRedCap), and expanded NR-NTN to Ka-band and sub-3 GHz for improved reliability [18], [19]. Release 19 (2025) integrates AI/ML-assisted mobility prediction, fast L1/L2 handovers, and enhanced LEO channel models for direct-to-handset satellite links [1]. Overall, the evolution from Release 13 to 19, and the upcoming Release 20 for 6G [20], illustrates the continuous advancement of RATs from LTE-based IoT toward intelligent,

unified, and energy-efficient 5G-Advanced and 6G systems connecting terrestrial and non-terrestrial networks.

2) *Non-3GPP RATs*: In parallel with 3GPP developments, non-3GPP standardization bodies such as IEEE, Bluetooth SIG and ETSI have advanced their own RATs, including Wi-Fi [21], Li-Fi [22], Bluetooth Low Energy (BLE) [23], LoRa [24], Ultra-Wideband (UWB) [25], Zigbee [26], Huawei NearLink [27], and ETSI DECT-2020 NR [4]. These technologies provide complementary connectivity to 3GPP systems, enabling seamless and flexible communication across heterogeneous network environments. For instance, Wi-Fi (IEEE 802.11) offers high throughput from hundreds of Mb/s to several Gb/s over short to medium ranges for indoor broadband and enterprise networks, while BLE (Bluetooth SIG) focuses on ultra-low-power, short-range links ideal for wearables and sensors. LoRa supports up to 15 km at very low data rates (50 kb/s), serving large-scale low-power IoT applications such as smart cities and agriculture. UWB provides centimeter-level localization and low-latency short-range communication for positioning, access control, and vehicular radar, whereas Zigbee offers moderate data rates with high energy efficiency for mesh-based industrial and home automation networks. Huawei NearLink, developed under China's SparkLink Alliance, combines Bluetooth and Wi-Fi like features with lower latency and higher energy efficiency for short-range control and vehicular use cases. ETSI DECT-2020 NR, the first non-3GPP technology recognized by the ITU as part of the IMT-2020 5G family, delivers reliable, license/license-exempt, time-synchronized links with deterministic latency and decentralized control for industrial and massive IoT deployments. Collectively, these non-3GPP RATs complement 3GPP systems by addressing diverse requirements in range, data rate, power efficiency, and deployment flexibility across consumer, industrial, and critical IoT domains.

However, these RATs were originally designed for distinct communication purposes, 3GPP systems prioritize wide-area mobility and reliability whereas non-3GPP technologies

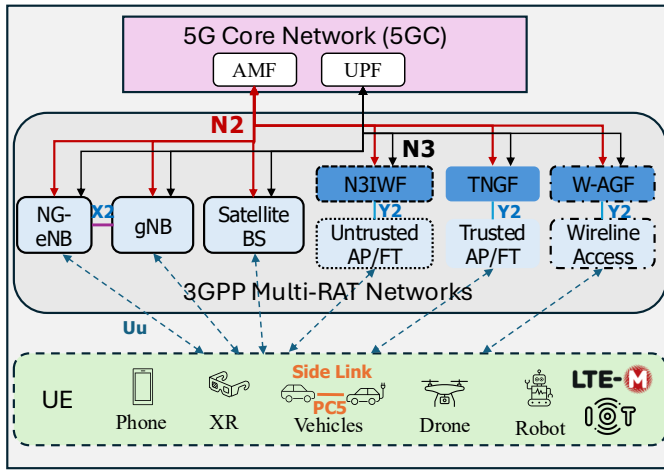


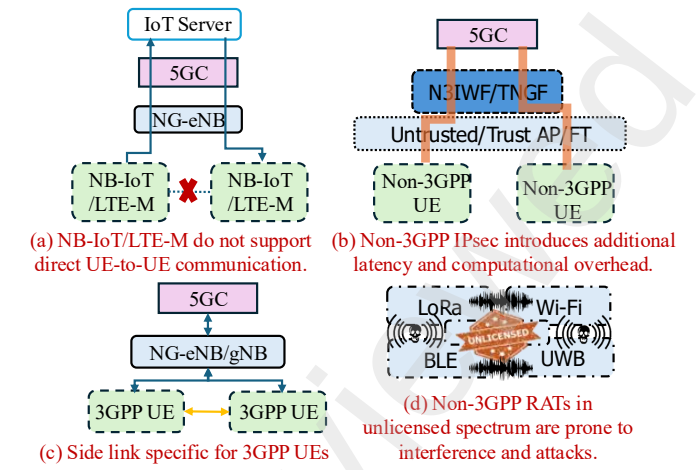
Fig. 2: 3GPP Multi-RAT Architecture and Limitations.

target short-range, low-power, and cost-efficient connectivity. Nevertheless, 3GPP-RATs provide global coverage but face limitations in indoor environments, whereas non-3GPP RATs, though flexible and inexpensive, often operate in unlicensed spectrum and are more susceptible to interference and security risks. The coexistence of these heterogeneous systems has led to fragmented architectures, isolated control domains, and inefficient spectrum utilization. Therefore, effective multi-RAT cooperation and integration are essential to unify their complementary strengths, enhance interoperability, and enable seamless, energy-efficient connectivity across diverse network environments.

B. Multi-RATs Cooperation and Integration

1) *3GPP RATs Cooperation and Integration:* As shown in Fig.2, within the 3GPP ecosystem, multiple generations of RATs coexist to serve diverse user groups, interconnected through standardized interfaces and service frameworks to ensure seamless interoperability [12]. For instance, LTE and NR are integrated via the NSA architecture, where LTE provides control-plane anchoring while NR delivers high-capacity user-plane data through the Xx and S1/N2 interfaces [28]. NB-IoT connects to the Evolved Packet Core (EPC) using the S1-U/S1-MME interfaces and can also interoperate with the 5GC via the N26 and N8 interfaces for seamless mobility. The sidelink (PC5) interface enables direct D2D communication in LTE and NR, supporting applications such as V2X and public safety. Additionally, NTN integrate with terrestrial 5G systems through the NG interface, extending connectivity via satellite access nodes connected to the 5GC. These standardized interconnections allow heterogeneous 3GPP RATs to operate cohesively within a unified architecture while supporting diverse service requirements.

2) *Non-3GPP RATs Cooperation and Integration:* Non-3GPP RAT cooperation enhances end-to-end connectivity by combining heterogeneous technologies to balance coverage, energy efficiency, and throughput. Commercial systems such as Apple AirDrop, Google Nearby Share, and Intel Unison



integrate BLE for device discovery with Wi-Fi Direct for data transfer, achieving fast and energy-efficient peer-to-peer (P2P) connectivity [29]. In low-power domains, LoRaBee and LoRaCP strengthen cross-technology connectivity between LoRa and ZigBee, where LoRaBee embeds specific bytes in LoRa packets to form detectable received signal strength patterns for one-hop control and synchronization, while LoRaCP establishes an out-of-band LoRaWAN control plane to enhance manageability and interference resilience [30], [31]. LoRaWAN and Wi-Fi integration extends this synergy by enabling Wi-Fi access points to host LoRaWAN servers, merging wide-area low-power sensing with high-throughput local access for cost-effective and spectrum coexistent IoT connectivity [32]. Recent physical layer cross technology communication schemes further enable direct LoRa and Wi-Fi interaction, where WiLo shapes 802.11g payloads into LoRa like chirps for gateway free communication, and LoFi allows Wi-Fi 6 receivers to reconstruct LoRa signals from OFDMA interference patterns using LDPC and neural reconstruction [33], [34]. Beyond low power systems, the DECT-2020 NR standard complements Wi-Fi in EWDC architectures by providing a secure, time-synchronized control link over dedicated spectrum, supporting deterministic communication for practical wireless time-sensitive networking [7]. Despite the advantages of non-3GPP RAT cooperation and integration in balancing coverage, energy efficiency, and cost, most of these technologies operate in unlicensed ISM bands, making them more vulnerable to security threats. Furthermore, their cooperative mechanisms often function in isolation, allowing a security breach in one RAT to propagate and compromise others.

3) *3GPP and Non-3GPP RATs Cooperation and Integration:* To mitigate these challenges, 3GPP Releases 13 and 14 introduced LTE and WLAN Aggregation (LWA) and LTE and WLAN integration with IPsec Tunnel (LWIP) to enhance multi-RAT connectivity [35]. In Release 13, LWA aggregates LTE and WLAN traffic at the PDCP layer via the

Xw interface and wireless termination node, enabling per-packet scheduling and efficient spectrum utilization, while LWIP integrates them at higher layers through IPsec tunnels to ensure compatibility with legacy WLAN deployments. Release 14 further introduced uplink aggregation, mobility continuity, and eNB assisted WLAN authentication, with results showing LWA's superior throughput and resource efficiency. Building on this evolution, 3GPP Release 16 introduced the Access Traffic Steering, Switching, and Splitting (ATSSS) framework to enable seamless multi-RATs integration under the 5GC through a unified multi-access session, leveraging multipath TCP (MPTCP) and multipath QUIC (MP-QUIC) to dynamically steer, switch, and split traffic across heterogeneous links for adaptive, policy driven multipath connectivity with improved throughput, latency, and reliability [36], [37]. Within ATSSS, WLAN can function as either a trusted or untrusted access network, connecting to the 5GC through a TNGF for trusted access or through a N3IWF for untrusted access, where UEs select the access type based on network advertisement and exchange NAS signaling and user plane data over IPsec tunnels with the corresponding gateway. Beyond 5G and WLAN integration, Mozny et al. [38] demonstrated a practical multi-RAT prototype combining NB-IoT and LoRaWAN for Smart City applications, where LoRaWAN offers cost-efficient primary connectivity and NB-IoT functions as a reliable fallback, thereby improving coverage, reliability, and overall QoS.

Recent research has further explored multi-RAT convergence by integrating cellular and non-cellular technologies within unified network architectures. For example, the 5G-CLARITY project demonstrated private 5G networks combining 5G NR, Wi-Fi, and LiFi to support multi-connectivity, high-precision localization, and cross-RAT slicing through the N3IWF for untrusted access [39]. In this design, non-3GPP RATs operate as independent systems connected to the 5GC via standardized interfaces, enabling centralized management. However, current 5GC support for inter-RAT mobility is largely limited to 3GPP systems such as LTE and NR via the N26 interface, while non-3GPP accesses provide only IP-level session continuity through IPsec-based gateways [40]. As a result, seamless radio-level handovers are not supported, and the latency introduced by IPsec encapsulation and gateway-based control signaling renders real-time mobility impractical for time-sensitive IIoT and URLLC applications [41]. Although the TNGF enables trusted access by requiring direct NAS signaling with the AMF and secure link-layer protection, it further increases configuration complexity and limits deployment flexibility in heterogeneous environments [6]. Moreover, both N3IWF and TNGF are primarily designed for individual non-3GPP RAT integration rather than coordinated multi-RAT operation, resulting in fragmented control, repeated authentication, and limited scalability. While these cooperation and integration mechanisms allow heterogeneous 3GPP and non-3GPP RATs to interoperate at the protocol and interface levels, they still lack unified management and orchestration capable of handling diverse QoS requirements and rapidly evolving RATs contexts. This gap motivates intelligent multi-RAT

management and orchestration, where AI/ML exploits cross-RAT observations and real-time control to optimize traffic steering, resource allocation, and mobility, thereby supporting 6G XR and IIoT services with extreme data rates, ultra-low latency, and high reliability through cost efficient use of complementary RAT capabilities.

C. Multi-RAT Management and Orchestration

1) Foundations of Multi-RAT Management Frameworks:

Building on 3GPP releases, various frameworks have been developed to unify heterogeneous Lite-RATs (e.g., Wi-Fi) under a common Big-RAT control architecture. In the LTE era, the EU SPEED-5G project introduced the extended Dynamic Spectrum Access (eDSA) concept, featuring a hierarchical MAC (High/Low-MAC) and distributed/centralized Radio Resource Management (dRRM/cRRM) to enable adaptive spectrum allocation and traffic steering across licensed and unlicensed bands [42], [43]. These concepts evolved into enhanced RRM (eRRM) and distributed scheduling frameworks, achieving fine-grained coordination between LTE and emerging NR layers; notably, Monteiro et al. [28] showed that distributed RRM combining UE-side association and BS-side scheduling can achieve near-centralized performance. In the 5G domain, the 3GPP multi-RAT dual connectivity framework, specified in TS 37.340 and validated through 5G-CARMEN trials, enables simultaneous use of LTE and 5G NR via master and secondary nodes to enhance user throughput and ensure real-time cross-RAT reliability [44]. Furthermore, Arendt et al. [45] developed an SDN-driven Multi-RAT broker for dynamic Wi-Fi and 5G spectrum sharing, while Famili et al. [46] realized slice-aware coordination in a cloud-native 5G/6G core. The emergence O-RAN architecture further enabled hierarchical control via non-RT rApps and near-RT xApps, where Habib et al. [47] applied hierarchical reinforcement learning (RL) for adaptive LTE–NR traffic steering. Within this evolving framework, AI and ML based methods are becoming key enablers for intelligent, secure, and context-aware RAT selection and coordination.

2) *AI-Enabled Multi-RAT Control and Optimization:* The evolution of multi-RAT control has advanced from static, spectrum-centric coordination toward AI-native, software-defined orchestration across radio and edge domains. Nguyen et al. [48] introduced network-assisted RL for distributed RAT selection, improving convergence and fairness across access networks. Alwarafy et al. [49] proposed DeepRAT, a hierarchical deep RL framework for joint RAT association and power control, while Allahham et al. [50] extended this to multi-agent coordination using a Multi-Agent Deep Deterministic Policy Gradient (MADDPG) approach for energy and QoS aware resource allocation. The aBRS framework [51] further integrates matching-game theory with double deep RL to enable bilateral, context-aware RAT negotiation. Building on these foundations, multi-agent RL has emerged as a dominant paradigm for autonomous control in O-RAN. More recently, LLM agents [52] have extended this paradigm by autonomously synthesizing interoperable APIs across heterogeneous network functions, supporting semantic-

level coordination across RATs. These developments mark the transition from rule-based spectrum coordination toward AI-native, agent-driven O-RAN ecosystems, enabling adaptive, interference-aware, and self-organizing control for intelligent 6G multi-RAT networks.

Despite significant progress in multi-RAT management and orchestration, existing frameworks primarily optimize RAT selection, resource allocation, and QoS provisioning, yet they do not resolve the fragmented control, authentication, and security models across heterogeneous 3GPP and non-3GPP RATs. Existing interworking mechanisms, such as N3IWF and TNGF, rely on IPsec to secure non-3GPP access, introducing substantial cryptographic overhead that increases latency and processing load incompatible with stringent 6G XR and IIoT requirements. Furthermore, each RAT maintains an independent secure connection to the 5GC, leading to repeated authentication, duplicated security contexts, and limited scalability. These limitations highlight the need for a unified multi-RAT architecture that offers a single control, authentication, and security domain while supporting high-throughput data transmission and seamless D2D communication. The next section introduces Uni-RAT, a unified multi-RAT framework designed to address these challenges.

D. Multi-RATs Security

Security and authentication remain critical for trusted interworking between 3GPP and non-3GPP RATs. As defined in 3GPP TS 33.501 and TS 23.316 [53], [54], the 5G Residential Gateway (5G-RG) functions as a UE within the 5GC, connecting via the Next Generation RAN (NG-RAN), the Wireline 5G Access Network (W-5GAN), or both, using 5G-AKA for 3GPP access and EAP-AKA' for non-3GPP access to ensure unified authentication and secure fixed-mobile convergence.

1) *5G-AKA and EAP-AKA' for Multi-RAT Security:* In 5G systems, mutual authentication between the UE and the core network is performed through coordinated interactions among several network functions. In the 5G-AKA procedure [53], the UE initiates registration by sending a Subscription Concealed Identifier (SUCI) to the AMF, which forwards an authentication request to the Authentication Server Function (AUSF). The AUSF retrieves the subscriber's long-term key and profile from the Unified Data Management (UDM) and returns an authentication vector containing a random challenge (RAND), an authentication token (AUTN), and a security anchor key (K_{SEAF}). The UE verifies AUTN and responds with RES*, which the AUSF validates before both entities derive session-specific keys for NAS and radio encryption. For non-3GPP access, the Extensible Authentication Protocol-AKA' (EAP-AKA') [55] follows a similar flow but encapsulates the exchange within an EAP tunnel carried over IPsec between the UE and the N3IWF or TNGF. The Authentication, Authorization, and Accounting (AAA) server acts as the EAP authenticator, interfacing with the AUSF/UDM to retrieve authentication vectors and validate responses. This unified authentication framework enables secure, privacy-preserving, and consistent key management across 3GPP and non-3GPP

access networks, ensuring trusted operation in heterogeneous multi-RAT environments.

2) *Untrusted Non-3GPP Access:* Over untrusted WLAN access, the UE first associates with the Wi-Fi network and obtains an IP address, then selects a N3IWF and initiates the IKEv2 procedure to establish an IKE Security Association (SA) [56]. The UE triggers an IKE_AUTH exchange without an AUTH payload to initiate an EAP-5G session, where NAS messages are encapsulated within IKEv2 signaling. UE authentication is performed using either EAP-AKA' or 5G-AKA. Upon successful authentication, the UE and N3IWF share a common N3IWF key from the AMF to the N3IWF and derived at the UE. IKEv2 then establishes a signaling IPsec SA providing both encryption and integrity protection, and the UE creates a TCP connection to carry subsequent NAS messages securely over this SA. For user-plane data transfer, one or more IPsec child SAs are established via IKEv2 during PDU session setup, each ensuring encryption and integrity protection.

3) *Trusted Non-3GPP Access:* Over trusted WLAN access, the UE associates with a Trusted WLAN Access Point (TNAP) and initiates an EAP procedure that triggers AAA signaling through the TNGF, which acts as a proxy to start an EAP-5G session [56]. The EAP-5G protocol encapsulates NAS messages between the UE and TNGF, while authentication is performed using either EAP-AKA' or 5G-AKA, with NAS signaling carried over IEEE 802.1X/EAPoL at Layer 2 and AAA (e.g., RADIUS) within the backhaul. After successful authentication, a common TNGF key is established and used to derive a TNAP key, which serves as the Pairwise Master Key (PMK) for executing the IEEE 802.11 four-way handshake to establish Layer 2 security. Once WLAN link-layer security and IP configuration are complete, the UE establishes a signaling IPsec SA with the TNGF via IKEv2 using the common TNGF key (typically with NULL encryption), followed by a TCP connection for reliable NAS message delivery. Subsequent user-plane traffic is then transported over secure IPsec tunnels. Despite 5G introducing unified authentication mechanisms such as 5G-AKA and EAP-AKA' for 3GPP and non-3GPP access, it relies on IPsec/IKEv2, adding computational overhead and latency. In addition, the 3GPP framework requires each RAT to establish its own session through the 5G Core, which prevents direct cross-RAT communication. While sidelink provides D2D connectivity exclusively for 3GPP UEs, commercial non-3GPP solutions such as Apple AirDrop offer low-latency direct communication but suffer from susceptibility to interference in unlicensed spectrum.

To address these limitations, the next section presents a DECT-2020 NR based Uni-RAT architecture and procedure, in which the DECT-2020 NR RAT operates over licensed spectrum and connects to the 5G Core to perform authentication and NAS establishment following 3GPP security procedures [57], [58] and ETSI TR 103 670 [59] interworking guidelines. In this design, DECT-2020 NR acts as the master RAT, coordinating Wi-Fi and other non-3GPP links to enable fast, secure, and direct cross-RAT connectivity.

III. UNI-RAT ARCHITECTURE AND SIGNALING

The Uni-RAT system integrates hardware-assisted UE and BS architectures with the 5GC, providing a secure and efficient foundation for coordinated multi-RAT operation. It complies with the standardized non-3GPP access framework defined in 3GPP TS 33.501 and TS 23.316 [53], [54], as well as the ETSI interworking model for DECT-2020 NR specified in TR 103 670 [59]. Conventional approaches interwork DECT-2020 NR with the 5GC via IPsec-protected N3IWF/TNGF gateways or fully 3GPP-capable relay devices, leading to higher deployment complexity and limited cross-RAT flexibility. In contrast, Uni-RAT employs lightweight WireGuard tunneling to secure both control-plane and data-plane communication, enabling efficient reuse of 3GPP security contexts for D2D and multi-hop scenarios while reducing cryptographic overhead and simplifying network management. Compliance with 3GPP networks is validated through integration with a standardized 5GC, as demonstrated in the following section.

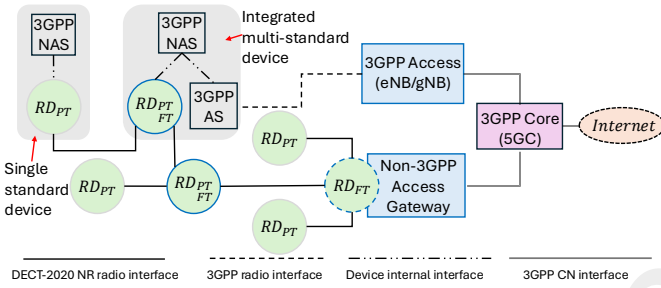


Fig. 3: ETSI DECT-2020 NR access to the 3GPP network.

A. Uni-RAT Architecture

Fig.3 illustrates the DECT-2020 NR architecture as a non-3GPP radio access network (RAN) for integration with the 3GPP system, as specified in ETSI TR103670 [59] and aligned with the 3GPP non-3GPP access framework defined in TS33.501 [53] and TS23.316 [54]. In this architecture, DECT-2020 NR radio devices (RDs) operating in Fixed Terminal (FT) mode can either access the 5GC through non-3GPP interworking gateways or function as 3GPP-capable nodes that connect to the 5GC via an eNB/gNB and relay 3GPP NAS signaling, thereby enabling non-3GPP RDs to access the 3GPP network. Portable Terminals (PTs) subsequently reuse the NAS security context shared by the FT to perform authentication and signaling with the 5GC. However, the former approach relies on N3IWF or TNGF with IPsec-based tunneling, requiring each RD to establish and maintain an independent secure connection to the 5GC, whereas the latter requires fully compliant 3GPP devices and base stations, resulting in increased deployment complexity and operational cost. Nevertheless, while DECT-2020 NR provides deterministic TDMA-based access well suited for IIoT and time-critical control, its peak data rate approximately only 1.2 Gbps remains significantly lower than that of modern Wi-Fi technologies (e.g., up to 9.6 Gbps for IEEE 802.11ax and up to 46 Gbps for IEEE

802.11be) [7]. This gap motivates a unified multi-RAT design in which DECT-2020 NR anchors secure and deterministic control-plane functions, while Wi-Fi delivers high-throughput data-plane communication, as realized in Uni-RAT.

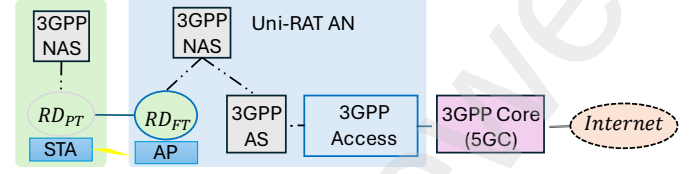


Fig. 4: Uni-RAT architecture.

Fig. 4 illustrates the Uni-RAT architecture, where a Uni-RAT Access Network (AN) interfaces directly with the 5GC and supports the required 3GPP Access Stratum (AS) and NAS procedures. Within the Uni-RAT, DECT-2020 NR RD_{FT} (BS side) and RD_{PT} (UE side) jointly form the Master-RAT that anchors NAS signaling, authentication, and security toward the 5GC, while Wi-Fi operates as a Secondary-RAT providing high-throughput data-plane connectivity. On the BS side, RD_{FT} is co-located with a Wi-Fi access point (AP) and exposes the established NAS security context through controlled interfaces, enabling Wi-Fi data delivery under Master-RAT security control. Similarly, on the UE side, RD_{PT} is co-located with Wi-Fi stations (STAs) and provides the NAS-derived security context for protected data-plane operation. This architecture enables secure local cross-RAT context sharing at both endpoints (FT/AP and PT/STA), while anchoring centralized 3GPP control-plane security in DECT-2020 NR, leveraging deterministic TDMA operation over licensed spectrum to ensure reliable and predictable control signaling.

B. Uni-RAT Signaling

Uni-RAT signaling aims to emulate 3GPP AS and NAS procedures through a trusted software module that implements standardized non-3GPP relay behavior, encompassing the DECT-2020 NR control-plane signaling, the Wi-Fi control-plane signaling, and the Uni-RAT user-plane over Wi-Fi.

1) *Uni-RAT control plane signaling with DECT-2020 NR:* Fig. 5 illustrates the Uni-RAT control-plane signaling, where the Uni-RAT AN emulates the trusted non-3GPP access network (TNAN), incorporating both TNAP and TNGF functions to enable fast and secure association over the licensed DECT-2020 NR spectrum. Unlike conventional TNAN deployments, Uni-RAT executes essential NGAP procedures, including initial UE message generation, security mode control, and UE context management, while securely relaying NAS signaling from the DECT-2020 NR side. At the same time, Uni-RAT replaces IPsec-based authentication with pre-installed WireGuard credentials to establish a secure and encrypted control-plane connection with the AMF. This WireGuard-based trust establishment effectively fulfills the authentication role of the TNGF, enabling verified and protected NAS signaling toward the 5GC with reduced setup latency and lower computational overhead. In the signaling procedure, DECT-2020 NR devices

first complete the native DECT association between the PT and FT. The PT then initiates the EAP-5G procedure with the Uni-RAT AN, which enables initial identity verification, key establishment, and secure transport of NAS messages toward the 5GC, supported by a minimal AAA function implemented within the Uni-RAT AN to satisfy 3GPP requirements. For the N2 interface, the Uni-RAT AN follows the gNB control-plane architecture by constructing NGAP signaling and encapsulating NAS messages into NGAP containers for delivery to the AMF. Once the N2 interface is established and NGAP signaling is active, the UE completes the 5G-AKA authentication procedure with the 5GC via the AMF, thereby finalizing secure control-plane establishment.

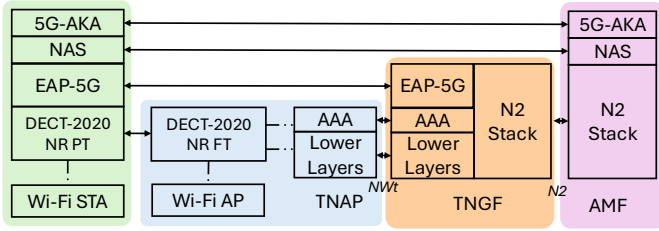


Fig. 5: Uni-RAT control plane signaling with DECT-2020 NR.

2) *Uni-RAT control plane signaling with Wi-Fi:* After successful 5G-AKA authentication over the DECT-2020 NR Master-RAT, Uni-RAT establishes control-plane signaling over the Wi-Fi Secondary-RAT, as illustrated in Fig. 6. First, the UE configures its Wi-Fi STA using network configuration parameters delivered from the Uni-RAT AN via NAS signaling over DECT-2020 NR. These parameters, including the SSID and access credentials, are conveyed using the SMS-over-NAS mechanism, enabling the UE Wi-Fi STA to associate with the Wi-Fi AP. Once the Wi-Fi association is completed, the Uni-RAT AN assigns the UE an outer IP address for data-plane data transmission. Using this outer IP address, the UE establishes a Layer-3 WireGuard (WG-L3) tunnel with the Uni-RAT AN, through which an inner IP address and a protected communication context are negotiated, enabling the UE to securely access the core network. All subsequent control-plane signaling between the UE and the 5GC is then carried over this NAS-authorized WireGuard tunnel, which serves as a lightweight and verifiable alternative to the IPsec-based tunneling employed in conventional TNAN deployments.

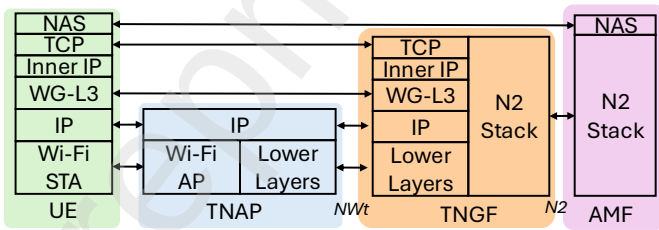


Fig. 6: Uni-RAT control plane signaling with Wi-Fi.

3) *Uni-RAT user plane signaling with Wi-Fi:* After completion of the control-plane procedures, the data plane is estab-

lished using the IP configuration and WG-L3 tunnel negotiated during control-plane signaling, including the assigned inner IP address, as illustrated in Fig. 7. On the UE side, user applications package data units (PDUs) are encapsulated with a generic routing encapsulation (GRE) header and an inner IP header, then protected by the WG-L3 tunnel before being transmitted over the Wi-Fi access link through the TNAP. The encrypted packets traverse the TNGF and are forwarded via the N3 interface toward the user plane function (UPF). Acting as the user-plane anchor, the UPF decapsulates the WG-L3 and GRE headers, thereby preserving QoS enforcement, session continuity, and routing transparency across heterogeneous access networks, and delivers the user PDUs to the target data network.

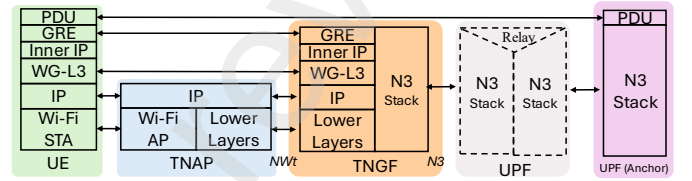


Fig. 7: Uni-RAT user plane signaling with Wi-Fi.

This section described the Uni-RAT architecture and signaling procedures, in which control-plane and user-plane functions are decoupled and coordinated across heterogeneous RATs. By anchoring 3GPP authentication, NAS signaling, and security in the DECT-2020 NR Master RAT and extending the established security context to Wi-Fi for high-throughput data delivery, Uni-RAT enables unified multi-RAT operation. Building on this foundation, the next section focuses on the concrete system realization of Uni-RAT.

IV. UNI-RAT SYSTEM DESIGN AND PROCEDURE

The Uni-RAT system integrates hardware-assisted UE and BS architectures with the 5GC, providing a secure and efficient foundation for coordinated multi-RAT operation. It complies with the standardized non-3GPP access framework defined in 3GPP TS 33.501 and TS 23.316 [53], [54], as well as the ETSI interworking model for DECT-2020 NR specified in TR 103 670 [59]. Conventional approaches interwork DECT-2020 NR with the 5GC via IPsec-protected N3IWF/TNGF gateways or fully 3GPP-capable relay devices, leading to higher deployment complexity and limited cross-RAT flexibility. In contrast, Uni-RAT employs lightweight WireGuard tunneling to secure both control-plane and data-plane communication, enabling efficient reuse of 3GPP security contexts for D2D and multi-hop scenarios while reducing cryptographic overhead and simplifying network management. Compliance with 3GPP networks is validated through integration with a standardized 5GC, as demonstrated in the following section.

A. Uni-RAT System Design

Fig. 8 illustrates the Uni-RAT system design, which comprises a Uni-RAT UE, a Uni-RAT BS acting as the TNAN, and corresponding enhancements in the 5GC. On the UE side,

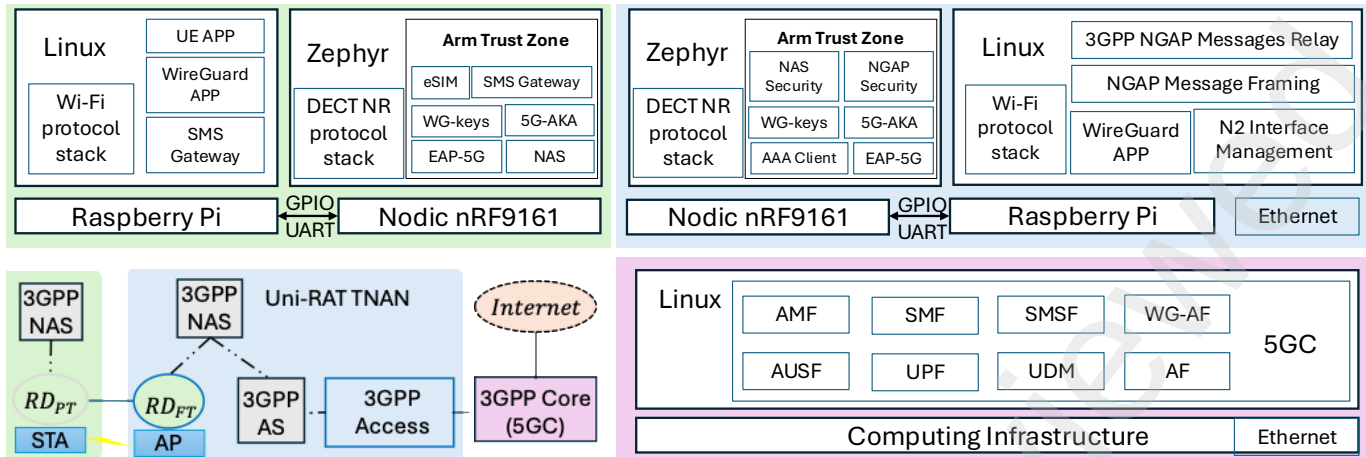


Fig. 8: Uni-RAT System Design.

Uni-RAT integrates a DECT-2020 NR PT implemented on the Nordic nRF9161 platform with ARM TrustZone support and a Linux-based Wi-Fi subsystem running on a Raspberry Pi. Security-critical functions, including eSIM handling, EAP-5G, 5G-AKA, NAS processing, and WireGuard key management, are executed inside the TrustZone, while the Linux domain hosts the Wi-Fi protocol stack, WireGuard application, and SMS gateway for flexible data-plane and control-plane interaction. On the BS side, the Uni-RAT BS combines a DECT-2020 NR FT on the nRF9161 with a co-located Wi-Fi AP on a Raspberry Pi, together emulating a TNAN that integrates both TNAP and TNGF functionalities. The BS implements trusted NAS and NGAP handling within TrustZone and relays 3GPP signaling to the 5GC via an N2 interface, while the Linux domain manages Wi-Fi access, WireGuard tunnels, and NGAP message framing. Within the 5GC, Uni-RAT extends standard core functions by incorporating enhanced SMS support and a dedicated WG-AF, which jointly enable secure key distribution, tunnel management, and rapid data-plane establishment across heterogeneous RATs.

1) *Uni-RAT UE*: The UE is logically partitioned into a non-secure UE stack and a secured UE stack, implemented across a Linux-based system and a Zephyr-based DECT-2020 NR platform, as illustrated in Fig. 9. The non-secure UE stack runs on a Raspberry Pi and hosts the Wi-Fi protocol stack, WireGuard application, SMS gateway, and user applications. This stack is responsible for assembling control-plane signaling, including SUCI, EAP-5G messages, NAS PDUs, and Wi-Fi configuration parameters, which are forwarded to the secure domain through GPIO or UART using the Secure Partition Manager (SPM) API. The secured UE stack executes within the Arm TrustZone on the Nordic nRF9161 and is responsible for all security-critical operations¹. These include eSIM services, SUCI generation, EAP-5G processing, 5G-AKA authentication, NAS key derivation and integrity protection, and WireGuard key management. Cryptographic primitives such as AES, ECC, hashing, true random number generation (TRNG), and ChaCha-based encryption are offloaded to the CryptoCell-310 accelerator, ensuring hardware-backed security and efficient cryptographic processing. With this hardware-assisted design, Uni-RAT confines authentication, NAS key derivation, and WireGuard key management to the DECT-2020 NR Master-RAT within ARM TrustZone and exposes them to Wi-Fi Secondary-RATs only through controlled service interfaces, enabling secure cross-RAT security-context sharing without exposing control-plane credentials.

2) *Uni-RAT BS*: The Uni-RAT BS functions as a TNAN, as illustrated in Fig. 10, and implements a control-plane state machine that manages UE access, NAS signaling, and NGAP interaction with the 5GC. The BS starts in the INIT state, where it initializes the TNAP components, ARM TrustZone environment and SPM APIs, configures the internal TNAP-TNGF pipeline, and establishes the N2 interface using SCTP and NGAP. Upon successful initialization, the system enters the IDLE/POLL state, continuously monitoring UE-side access traffic and control messages from the AMF. When a UE

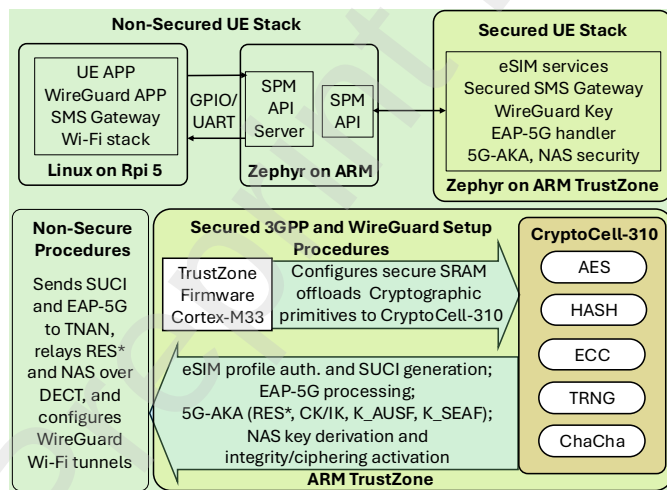


Fig. 9: Uni-RAT UE Design.

¹https://docs.nordicsemi.com/bundle/ps_nrf9151/page/cryptocell.html

TABLE I: Uni-RAT 5GC API Summary

5GC	API Description
AMF	CreateUeContext: create the AMF UE context and process the initial NAS message. SendUplinkNas: forward UE-originated NAS from the Uni-RAT BS to the AMF. PushDownlinkNas: deliver NAS/NGAP messages from the AMF to the Uni-RAT BS. ReleaseUeContext: deregister the UE and release associated N2 resources.
SMSF	SubmitMoSms: handle a MO SMS sent by the UE via the Uni-RAT BS and AMF. SubmitMtSms: deliver a MT SMS to the UE, including Uni-RAT WireGuard bootstrap when required. ReportSmsStatus: report SMS delivery outcomes and acknowledgment status.
WG-AF	RequestWgTunnel: allocate WireGuard keys, tunnel endpoints, and configuration parameters. UpdateWgTunnel: rekey, modify, or delete an existing WireGuard tunnel instance. GetWgStatus: retrieve the operational state of a WireGuard tunnel.
UDM	GetAuthData: provide authentication vectors for 5G-AKA or EAP-5G procedures. GetWgProfile: return the subscriber's WireGuard policy and allowed peer configuration. ReportWgUsage: update subscriber WireGuard usage records and policy state.

attach is detected, the BS transitions to **HANDLE ATTACH**, where a UE context is created, authentication credentials (EAP-5G/5G-AKA) are retrieved from TrustZone, and an InitialUEMessage is sent to the AMF. During operation, **HANDLE UPLINK** and **HANDLE DOWNLINK** states process UE-originated and AMF-originated NAS/NGAP messages, respectively, advancing the UE procedure state as required. Finally, the **CLEANUP** state securely releases UE contexts and erases NAS and keying material from TrustZone upon UE detachment or AMF-triggered release.

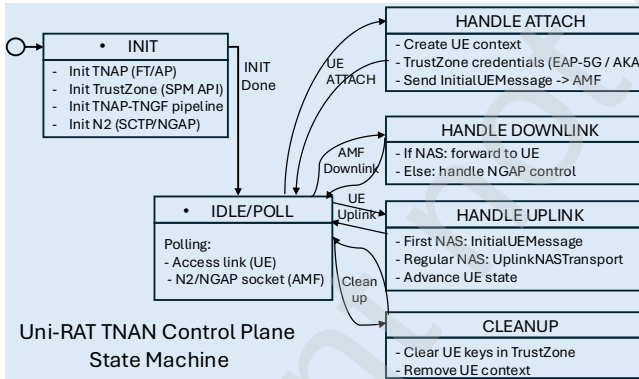


Fig. 10: Uni-RAT BS (TNAN) Design.

3) *Uni-RAT 5GC*: The 5GC adopts a service-based architecture (SBA), in which Uni-RAT extends existing network functions with a set of dedicated APIs to support unified multi-RAT authentication, signaling, and secure data-plane establishment. As summarized in Table I, the AMF exposes APIs for UE context management and NAS/NGAP message exchange with the Uni-RAT BS, while the SMSF is enhanced to support mobile-originated (MO) and mobile-terminated (MT) SMS, including SMS-over-NAS-based WireGuard bootstrap signaling. A dedicated WG-AF is introduced to allocate, manage, and monitor WireGuard tunnels, and the UDM is

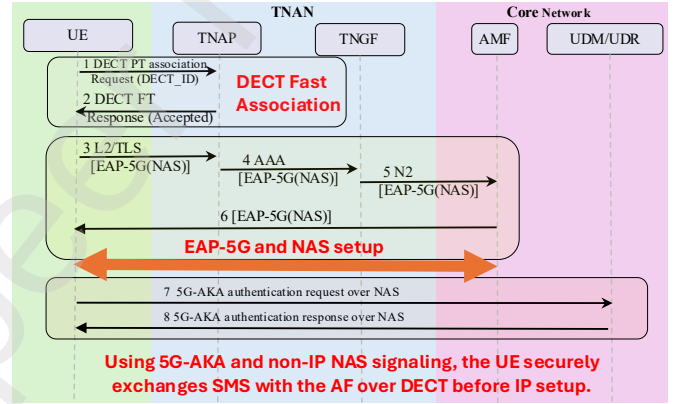


Fig. 11: UE association and authentication.

extended to provide authentication vectors and per-subscriber WireGuard profiles. Together, these SBA interfaces enable secure, scalable, and tightly coordinated control-plane and data-plane operations across heterogeneous RATs in Uni-RAT. With the integrated UE, BS, and 5GC design, the Uni-RAT system procedures encompass UE association and authentication, intra-cell D2D WireGuard tunnel establishment, and inter-cell multi-hop WireGuard setup coordinated by the WG-AF, as presented in the following subsections.

B. Uni-RAT Procedures

1) *UE Association and Authentication*: Fig. 11 illustrates the UE association and authentication procedure based on DECT-2020 NR control-plane signaling. The process starts with a native DECT-2020 NR association between the UE-side RD_{PT} and the BS-side RD_{FT} within the Uni-RAT AN. During this phase, the UE completes DECT link establishment and device identification, providing a deterministic and trusted access channel over licensed spectrum. After DECT association, the UE initiates the EAP-5G procedure over the DECT-2020 NR access. EAP-5G messages are encapsulated in NAS

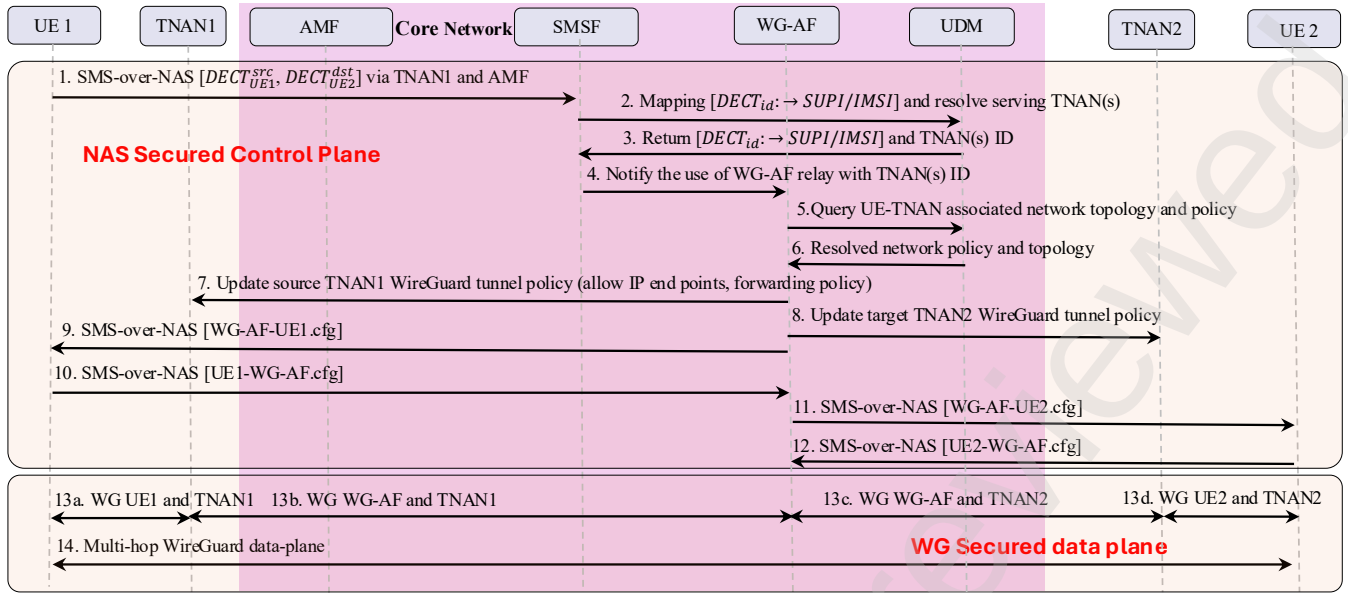


Fig. 12: Inter-cell multi-hop WireGuard setup via WG-AF.

signaling and forwarded by the Uni-RAT AN, which emulates a TNAN, toward the AMF via the N2 interface. The Uni-RAT AN executes the required NGAP procedures and relays EAP-5G(NAS) messages to the 5GC, supported by a minimal AAA function to satisfy 3GPP authentication requirements. Successful EAP-5G completion establishes an initial NAS security context between the UE and the 5GC. Subsequently, the UE performs the 5G-AKA authentication procedure over the established NAS connection. Authentication challenges and responses are exchanged between the UE and the AMF, resulting in mutual authentication and derivation of NAS encryption and integrity keys. During this phase, Uni-RAT enables secure exchange of control information, including SMS-over-NAS messages with the SMS-AF, entirely over the DECT-2020 NR Master-RAT before IP-layer setup.

In this case, UE1 initiates the procedure by sending an SMS-over-NAS message containing the source and destination DECT identifiers ($DECT_{src}$, $DECT_{dst}$) via its serving TNAN and the AMF. The 5GC resolves the corresponding 3GPP identities and determines the serving TNANs of both UEs using the UDM. Upon detecting an inter-cell scenario, the SMSF invokes the WG-AF to coordinate multi-hop connectivity across TNANs. The WG-AF queries the UDM for UE-specific network topology and policy information, then derives forwarding rules and tunnel policies for the source TNAN, target TNAN, and intermediate WG-AF relay. WireGuard configuration parameters are distributed to UE₁, UE₂, and the involved TNANs via SMS-over-NAS. Subsequently, WireGuard tunnels are established between UE₁ and its serving TNAN, between TNANs and the WG-AF, and between the WG-AF and the target TNAN, finally reaching UE₂. Subsequently, WireGuard tunnels are established between UE₁ and its serving TNAN, between each TNAN and the WG-AF, and between the WG-AF and the target TNAN, ultimately reaching UE₂. These authenticated and encrypted tunnel segments are chained at the WG-AF, forming a secure multi-hop WireGuard data plane across cells while preserving a strict separation between the NAS-secured control plane and the WireGuard-protected data plane.

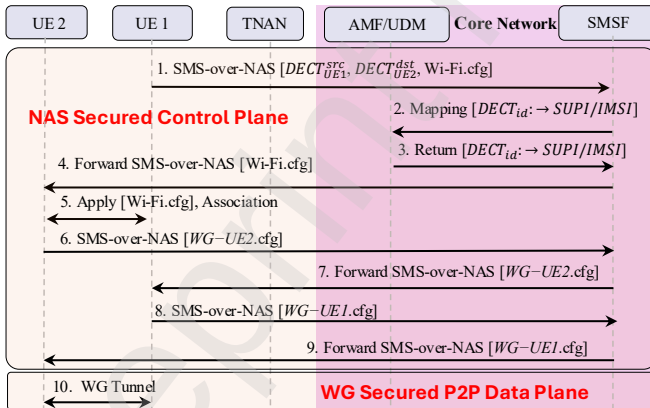


Fig. 13: Intra-cell D2D WireGuard setup.

2) *Inter-cell multi-hop WireGuard setup via WG-AF:* Fig. 12 illustrates the inter-cell multi-hop WireGuard setup procedure, which is triggered when UE₁ and UE₂ are located

3) *Intra-cell D2D WireGuard setup:* Fig.13 illustrates the intra-cell D2D WireGuard setup procedure in Uni-RAT. Following successful NAS security establishment over the DECT-2020 NR Master-RAT, the Uni-RAT AN coordinates D2D connectivity by distributing Wi-Fi and WireGuard configuration parameters via SMS-over-NAS. In this procedure, Uni-RAT UE1 discovers UE2 within the same cell and initiates a D2D connection by sending an SMS-over-NAS message via the SMSF, carrying the $DECT_{id}$ of both UEs and the required

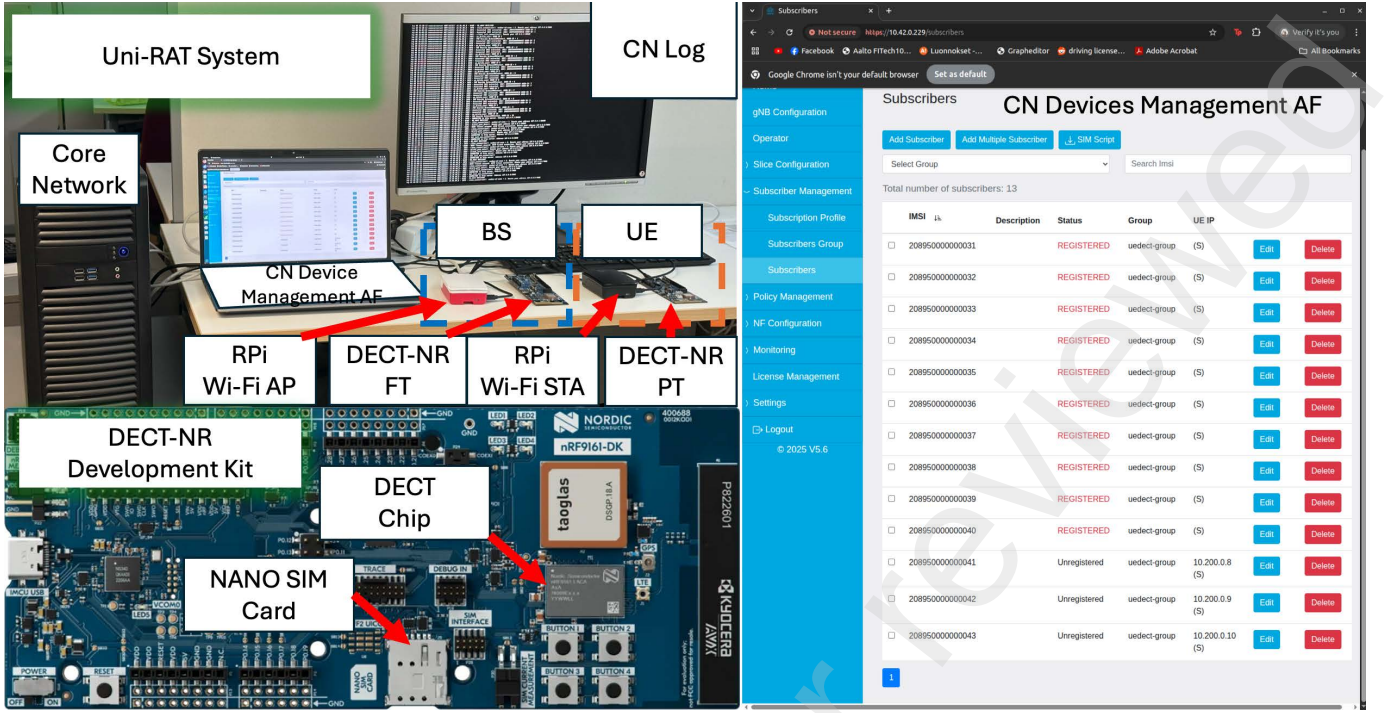


Fig. 14: Uni-RAT System Setup.

Wi-Fi configuration parameters (e.g., SSID and credentials). The 5GC resolves the $DECT_{id}$ values to 3GPP identities through the AMF and UDM and verifies that both UEs are served by the same cell. Upon validation, the SMSF forwards the configuration to UE_2 over the NAS-secured control plane. After Wi-Fi association with the Uni-RAT TNAN, the UEs exchange additional SMS-over-NAS messages to distribute WireGuard configuration parameters, including peer keys and tunnel endpoints. Following mutual confirmation, UE_1 and UE_2 establish a direct WireGuard tunnel, forming a secure point-to-point D2D data plane. This procedure enables centrally controlled, low-latency D2D communication while offloading traffic from the UPF and preserving unified 3GPP security control. Based on the Uni-RAT architecture and signaling procedures described above, the following section presents the concrete system implementation and experimental evaluation of Uni-RAT.

V. UNI-RAT IMPLEMENTATION AND EVALUATION

This section details the hardware and software realization of the Uni-RAT UE, BS, and 5GC enhancements, followed by protocol-level verification using captured NAS and WireGuard signaling. It further evaluates data-plane performance in D2D and multi-hop scenarios, as well as control-plane scalability under large-scale UE and BS registration loads.

A. Uni-RAT System Setup

Fig. 14 illustrates the Uni-RAT system setup used for implementation and evaluation. The testbed comprises a dedicated server running the CumuCore 5GC and a management laptop connected to the core network for operating the

TABLE II: Uni-RAT System Setup Parameters

Device	Parameter / OS	Setting / Version
DECT-2020 NR	Operating Frequency Channel Bandwidth Subcarrier Spacing Frame Duration Transmit Power Modulation Scheme	1888.704 MHz 1.728 MHz 27 kHz 10 ms 13 dBm QPSK
Wi-Fi AP TNAN1	Channel Number Center Frequency Channel Bandwidth Transmit Power	Channel 1 2412 MHz 20 MHz 31 dBm
Wi-Fi AP TNAN2	Channel Number Center Frequency Channel Bandwidth Transmit Power	Channel 11 2462 MHz 20 MHz 31 dBm
Nordic nRF9161 Development Kit	Modem Tick Rate UART Baud Rate UART Buffer Size SDK Zephyr OS	69.12 MHz 115200 200 Bytes v3.0.2 v4.0.99
RPI AP RPI STA	Debian Debian	11.11 12.11
5GC Server 5GC Laptop	Ubuntu Ubuntu	24.04.3 LTS 22.04.5 LTS

graphical user interface (GUI). The GUI supports UE and base station provisioning, subscriber profile configuration, and access control based on context information maintained in the UDM. Through predefined 5GC APIs, the GUI enables real-time visualization and management of registered DECT-2020 NR UEs and associated base stations. As shown in the management interface, the device management AF groups connected DECT-2020 NR devices under the *uedectgroup*, displays their corresponding IMSI information, and allows operators to manage or revoke device access. The appearance of both DECT UEs and base stations in the GUI confirms successful authentication and registration with the 5GC.

Table II summarizes the radio configurations, platforms, and operating systems used in the Uni-RAT experiments. The DECT-2020 NR radio operates at a carrier frequency of 1888.704 MHz with a bandwidth of 1.728 MHz, 27 kHz subcarrier spacing, QPSK modulation with coding rate 1/2, and a transmit power of 13 dBm, providing a low-latency and reliable control-plane access link. Two Wi-Fi access points are configured on non-overlapping channels, with one operating on Channel 1 (2412 MHz) and the other on Channel 11 (2462 MHz); both use a 20 MHz channel bandwidth and a transmit power of 31 dBm to support data-plane transmission. The DECT-2020 NR radios are implemented on Nordic nRF9161 platforms running nRF Connect SDK v3.0.2 and ZephyrOS v4.0.99, while Uni-RAT BS and UE functions are hosted on Raspberry Pi platforms, with the BS deployed on an RPi4 under Debian 11.11 to enable low-level Wi-Fi MAC configuration required for controlled experiments, and the UE deployed on an RPi5 under Debian 12.11 using a Broadcom Wi-Fi chipset whose closed MAC layer limits fine-grained channel control, reflecting current hardware constraints in practical evaluations. Respectively. The 5GC is deployed on an Ubuntu 24.04.3 LTS server with management and visualization supported by an Ubuntu 22.04.5 LTS laptop, forming a complete E2E Uni-RAT testbed for both control and data plane evaluation.

B. Uni-RAT System Implementation

1) *Uni-RAT UE and BS Implementation:* Fig. 15 present the Uni-RAT UE and BS TrustZone during debugging logs. The logs first confirm secure boot and initialization of the nRF9161 platform, followed by UE context creation using the provisioned SUPI and construction of the NAS registration request PDU, whose encoding is verified against the expected 3GPP format. During the 5G-AKA procedure, the TrustZone logs demonstrate that all security-critical NAS processing strictly follows the authentication flow defined in 3GPP TS33.501 [53] and is executed entirely within the secure world. Specifically, upon reception of the NAS authentication request (AUTH-REQ), the TrustZone parses the NAS header and key set identifier (KSI), extracts the anti-bidding down between architectures (ABBA) parameter used to bind authentication to the 5G security architecture, and processes the 128-bit RAND and 128-bit AUTN values. The AUTN, comprising the concealed sequence number ($SQN \oplus AK$), authentication

management field (AMF), and message authentication code (MAC), enables network authentication and protection against replay, impersonation, and downgrade attacks. Using these inputs, the TrustZone verifies the MAC, recovers the SQN, and computes the 5G-AKA response (RES^*), which is encapsulated in the NAS Authentication Response PDU. In parallel, the TrustZone derives the NAS security context, including the NAS encryption key (K_{enc}) and integrity key (K_{int}), thereby securing all subsequent NAS signaling between the UE and the 5GC. After successful authentication, NAS messages are protected using the NEA2 algorithm specified in TS 33.501, which is based on 128-bit AES in counter (CTR) mode.

For WireGuard integration, the TrustZone logs confirm that each UE generates a static public-private key pair using the PSA Crypto API. The private key is created, stored, and accessed exclusively within the TrustZone and is never exposed to the non-secure domain, while the corresponding public key is safely exported for distribution to authorized peers via the SMS-over-NAS control channel. The TrustZone also securely stores public keys provisioned by the 5GC, enabling authenticated tunnel establishment between the UE, TNAN, and core network. This design replaces the certificate-based trust model of IPsec and TNGF with a lightweight, hardware-protected authentication mechanism. During data-plane setup, WireGuard performs a Noise-based handshake to derive ephemeral session keys from the static key pairs, providing mutual authentication, forward secrecy, and encrypted data-plane communication. Overall, this implementation anchors authentication, NAS security, and WireGuard key management within hardware-isolated TrustZone environments, enabling secure control-plane signaling and protected data-plane transmission for Uni-RAT in 6G.

2) *Uni-RAT 5GC Implementation:* Fig.16 illustrates the Uni-RAT system procedures observed during 5GC operation. The DECT-2020 NR association phase is captured using a custom passive sniffer developed in our prior work [7]. This sniffer leverages the nRF9161 PHY-layer API to monitor DECT-2020 channels, extract raw PHY frames, remove PHY headers, and forward hexadecimal payloads via a local UDP pipeline to Wireshark, where a custom dissector decodes DECT-2020 MAC and SDU fields for detailed inspection of beacons, association signaling, and data PDUs.

Following DECT association, the Uni-RAT UE initiates the 5G registration and authentication procedure by transmitting a NAS message carrying an EAP-5G payload over the DECT-2020 NR link to the TNAN (IP 10.42.1.1). In parallel, the TNAN establishes the N2 control-plane connection with the AMF (IP 10.42.1.27) using SCTP, beginning with the standard four-way handshake. The exchange of *INIT* and *INIT-ACK* messages negotiates transport parameters, followed by *COOKIE-ECHO* and *COOKIE-ACK* messages that enable stateless association setup and protect against spoofing and resource-exhaustion attacks. Once the association is established, SCTP *Selective Acknowledgement* (*SACK*) messages ensure reliable, ordered, and loss-resilient transport of NGAP signaling. Over the established SCTP association, the TNAN

```

*** Booting nRF Connect SDK v3.0.2-89ba1294ac9b ***
*** Using Zephyr OS v4.0.99-f791c49f492c ***

=== UE context init ===
Supi: 208950000000031

=== Construct Registration Request PDU ===
Actual REG-REQ (25): 7e004171000d0102f8592143000000000000132e0480200000
Expect REG-REQ (25): 7e004171000d0102f8592143000000000000132e0480200000
[PASS] Registration Request PDU matches expected

=== Process Authentication Request PDU ===
Input AUTH-REQ (42): 7e005602020000215622ffab4ac717d18934ff154751803d20104b411775fe868000445aa57c452d2a22
NAS header: 7e 00 56
NAS KSI: 0x02
ABBA: 00 00
RAND: 56 22 ff ab 4a c7 17 d1 89 34 ff 15 47 51 80 3d
AUTN: 4b 41 17 75 fe 86 80 00 44 5a a5 7c 45 2d 2a 22
AMF: 80 00
MAC: 44 5a a5 7c 45 2d 2a 22
SQN xor AK: 4b 41 17 75 fe 86
SQN: 00 00 00 00 00 40
IK: 9c 1d de 76 25 8c 2f 71 51 bb 8d 66 8b 66 9a 11
CK: 54 64 02 45 d1 74 47 de 2b 6e f0 ad c0 66 2d 44
AK: 4b 41 17 75 fe c6
RES: 60 6e 7a e4 29 b2 63 77
RESSTAR: 79 5f 02 b9 06 c3 7f 4f 69 0c ab e1 1f ea 83 09
Kenc: c8 70 4b ff 84 ba d9 0a fa 21 6a 5b fa 6d 93 cf
Kint: 5f 24 bf 6c 56 d3 2e aa 0f c5 77 cb f4 80 3d 90

=== Construct Authentication Response PDU ===
Actual AUTH-RES (21): 7e00572d10795f02b906c37f4f690cabe11fea8309
Expect AUTH-RES (21): 7e00572d10795f02b906c37f4f690cabe11fea8309
[PASS] Authentication Response PDU matches expected

=== NEA2 Encrypt/Decrypt roundtrip ===
nea2_encrypt(plain->cipher)
Plain AUTH-RES (21): 7e00572d10795f02b906c37f4f690cabe11fea8309
Cipher AUTH-RES (21): f83e002fe67bd1d0d65cbf37e5532419614977e045
nea2_decrypt(cipher->plain)
Recover AUTH-RES (21): 7e00572d10795f02b906c37f4f690cabe11fea8309
[PASS] Recovered PDU match original PDU

=== WireGuard keypair sanity (PSA, private key in Secure) ===

=== WG DEBUG: export private key (DO NOT SHIP) ===
WG Private Key (exported) (32): a8 22 cd 20 d0 74 05 c0 88 4d 4b d1 3c 23 28 8c 8d 78 f7 fe e2 e2 fb d6 88 1c dd 7c 83 60 4e 50
WG Public Key (32): bc 97 0d f9 a2 9b 02 6c 7c ee 5d b0 82 23 96 e0 23 e6 53 13 37 20 05 b3 48 dc 36 39 4c 74 09 08

[SUMMARY] 3/3 passed

```

Fig. 15: Uni-RAT UE TrustZone authentication, NAS security processing, and WireGuard key management log.

performs the NGAP Setup procedure and relays UE-related NGAP and NAS signaling, including the *InitialUEMessage* carrying the NAS Registration Request. Subsequent downlink and uplink NAS transport messages support the EAP-5G and 5G-AKA authentication exchange, security mode control, and initial UE context setup. The observed signaling follows the standardized 5G registration and authentication workflow, with NAS messages encapsulated in NGAP and transported over SCTP, as confirmed by the captured NGAP/NAS traces. The complete Wireshark trace files are publicly available in the project GitHub repository². Correct decoding of DECT-2020 NR packets requires the custom Wireshark dissector developed in [7], which is also provided in the repository and must be installed to enable full protocol-level analysis.

In real-world deployments, however, the NGAP setup and UE authentication signaling exchanged between the TNAN and the AMF cannot be directly captured, as the N2 interface is fully encrypted using WireGuard with pre-shared keys securely stored within the ARM TrustZone. Consequently,

such control-plane signaling is observable only during controlled debugging and development scenarios. In addition, the SMSF and WG-AF components used in this work are part of the Cumucore 5G Core and require appropriate licenses for deployment and evaluation.

3) *Uni-RAT WireGuard Implementation*: Fig.17 illustrates the Uni-RAT WireGuard configuration and data-plane operation corresponding to the SMS-assisted intra-cell D2D setup described in Fig.13. As shown in Fig.17a and Fig.17b, UE1 and UE2 establish a direct WireGuard tunnel after exchanging configuration parameters via SMS-over-NAS, with each UE maintaining a WireGuard interface whose policy is bound to the peer's internal tunnel IP address. In this configuration, the *endpoint* IP corresponds to the peer's Wi-Fi interface address and port, which are used to transport encrypted WireGuard packets over the local Wi-Fi link, while the *AllowedIPs* field specifies the peer's virtual tunnel IP address whose traffic is routed through the secure WireGuard interface. This separation between transport-layer addressing and tunnel-layer addressing enables stable D2D communication and policy enforcement independent of the underlying access network. The Wireshark

²<https://version.aalto.fi/gitlab/dangy1/uni-rat.git>

[dect_nr wg (sctp.chunk_type != 4 && sctp.port == 38412 && sctp.chunk_type != 5)]						
No.	Time	Source	Destination	Protocol	Length	Info
8336	46.117731...	127.0.0.1	127.0.0.1	dect_nr	84	33440 → 8091 Len=42
8337	46.127622...	127.0.0.1	127.0.0.1	dect_nr	84	33440 → 8091 Len=42
8338	46.137592...	127.0.0.1	127.0.0.1	dect_nr	84	33440 → 8091 Len=42
8339	46.142414...	127.0.0.1	127.0.0.1	dect_nr	51	33440 → 8091 Len=9
8374	46.190746...	127.0.0.1	127.0.0.1	dect_nr	84	33440 → 8091 Len=42
8375	46.192405...	127.0.0.1	127.0.0.1	dect_nr	51	33440 → 8091 Len=9
8616	47.691555...	10.42.1.1	10.42.1.27	SCTP	82	INIT
8617	47.692160...	10.42.1.27	10.42.1.1	SCTP	306	INIT_ACK
8620	47.692207...	10.42.1.1	10.42.1.27	SCTP	278	COOKIE_ECHO
8622	47.692484...	10.42.1.27	10.42.1.1	SCTP	60	COOKIE_ACK
8624	47.692561...	10.42.1.1	10.42.1.27	NGAP	106	NGSetupRequest
8626	47.692728...	10.42.1.27	10.42.1.1	SCTP	62	SACK (Ack=0, Arwnd=106452)
8628	47.693330...	10.42.1.27	10.42.1.1	NGAP	146	NGSetupResponse
8632	47.693346...	10.42.1.1	10.42.1.27	SCTP	62	SACK (Ack=0, Arwnd=106412)
9703	53.555939...	10.42.1.1	10.42.1.27	NGAP/NAS...	134	InitialUEMessage, Registration request [RRCEstablishmentCause=mo-Signalling]
9749	53.630227...	10.42.1.27	10.42.1.1	NGAP/NAS...	150	SACK (Ack=1, Arwnd=106496), DownlinkNASTransport, Authentication request
9752	53.636672...	10.42.1.1	10.42.1.27	NGAP/NAS...	146	SACK (Ack=1, Arwnd=106496), UplinkNASTransport, Authentication response
9754	53.656464...	10.42.1.27	10.42.1.1	NGAP/NAS...	122	SACK (Ack=2, Arwnd=106496), DownlinkNASTransport, Security mode command
9758	53.682556...	10.42.1.1	10.42.1.27	NGAP/NAS...	134	SACK (Ack=2, Arwnd=106496), UplinkNASTransport
9765	53.879943...	10.42.1.27	10.42.1.1	NGAP/NAS...	230	SACK (Ack=3, Arwnd=106496), InitialContextSetupRequest
9831	53.896084...	10.42.1.1	10.42.1.27	NGAP	102	SACK (Ack=3, Arwnd=106496), InitialContextSetupResponse
9832	53.896132...	10.42.1.1	10.42.1.27	NGAP/NAS...	118	UplinkNASTransport
9834	53.896382...	10.42.1.27	10.42.1.1	SCTP	62	SACK (Ack=5, Arwnd=106418)
9836	53.896543...	10.42.1.27	10.42.1.1	NGAP	90	DownlinkNASTransport
9837	54.103716...	10.42.1.1	10.42.1.27	SCTP	62	SACK (Ack=4, Arwnd=106496)
9912	54.630543...	10.42.1.1	10.42.1.27	NGAP/NAS...	134	InitialUEMessage, Registration request [RRCEstablishmentCause=mo-Signalling]
9916	54.641917...	10.42.1.27	10.42.1.1	NGAP/NAS...	150	SACK (Ack=6, Arwnd=106496), DownlinkNASTransport, Authentication request
9920	54.643225...	10.42.1.1	10.42.1.27	NGAP/NAS...	146	SACK (Ack=5, Arwnd=106496), UplinkNASTransport, Authentication response
9924	54.648621...	10.42.1.27	10.42.1.1	NGAP/NAS...	122	SACK (Ack=7, Arwnd=106496), DownlinkNASTransport, Security mode command
9925	54.651826...	10.42.1.1	10.42.1.27	NGAP/NAS...	134	SACK (Ack=6, Arwnd=106496), UplinkNASTransport
9929	54.680049...	10.42.1.27	10.42.1.1	NGAP/NAS...	230	SACK (Ack=8, Arwnd=106496), InitialContextSetupRequest
9933	54.680908...	10.42.1.1	10.42.1.27	NGAP	102	SACK (Ack=7, Arwnd=106496), InitialContextSetupResponse
9936	54.680941...	10.42.1.27	10.42.1.1	NGAP/NAS...	118	UplinkNASTransport
9940	54.681182...	10.42.1.27	10.42.1.1	SCTP	62	SACK (Ack=10, Arwnd=106418)
9944	54.681419...	10.42.1.27	10.42.1.1	NGAP	90	DownlinkNASTransport
10018	54.883679...	10.42.1.1	10.42.1.27	SCTP	62	SACK (Ack=8, Arwnd=106496)

Fig. 16: Wireshark packet capture and signaling procedure of the Uni-RAT system.

capture in Fig. 17c illustrates WireGuard data-plane communication between two Uni-RAT UEs over the Wi-Fi link. Each packet labeled Transport Data represents an encrypted WireGuard message carried over UDP, where the receiver field enables each endpoint to map the packet to the correct peer and cryptographic context, the monotonically increasing counter enforces packet ordering and replay protection and confirms session state synchronization between the UEs, and the datalen field indicates the size of the encrypted payload, with larger values corresponding to encapsulated application data and smaller values reflecting control or acknowledgment traffic. The observed bidirectional exchange of packets with valid receiver identifiers, increasing counters, and encrypted payloads confirms successful WireGuard handshake completion and demonstrates that Uni-RAT provides E2E confidentiality, integrity, and replay protection while transparently operating over Wi-Fi and offloading traffic from the 5GC UPF.

AppendixA presents the multi-hop WireGuard configurations for Uni-RAT inter-cell communication. Two UEs connect to different Uni-RAT TNANs, which establish WireGuard tunnels to a shared relay node. Each TNAN maintains peers toward its UE and the relay and applies routing rules for remote-cell traffic. The relay forwards encrypted traffic between TNANs using IP forwarding and NAT, enabling inter-cell multi-hop communication without involving the 5GC UPF. Listings6–7 show the corresponding nftables configuration. Dynamic key management and topology adaptation via the WG-AF are left for future work.

C. Uni-RAT System Evaluation

1) *Uni-RAT UE and TNAN Evaluation:* To evaluate the proposed Uni-RAT system in terms of throughput, packet loss rate, latency, and jitter, three scenarios were designed and tested: intra-cell D2D communication, where two UEs communicate directly within the same cell; a one-hop relaying scenario, where two UEs in the same cell cannot communicate directly due to distance and the TNAN acts as a relay; and an inter-cell scenario, where the two UEs are located in different cells and communication is relayed across cells via the TNAN.

Fig.18 presents the throughput performance measured using Iperf3 under different target bandwidth configurations for all scenarios. The results show that intra-cell D2D communication consistently achieves the highest throughput, closely tracking the target bandwidth until reaching the practical wireless capacity limit. This is expected, as D2D communication benefits from direct connectivity, involving only a single WireGuard tunnel, and is supported by UE devices equipped with more capable Wi-Fi chipsets compared to the TNAN AP. In contrast, the one-hop and inter-cell multi-hop scenarios exhibit comparable throughput and saturate at similar levels. Both scenarios rely on the TNAN access point operating in the 2.4 GHz ISM band with a 20 MHz channel bandwidth, implemented on a Raspberry Pi 4, which lacks support for newer Wi-Fi protocols and advanced PHY features. As a result, throughput in these scenarios is constrained by the practical capacity of the wireless backhaul, rather than by the Uni-RAT architecture itself. These results indicate that the performance

```

dang@raspberrypi:~/dect-gnbsim/build $ sudo wg
interface: wg0
  public key: YBgIvD3LeiLQC/35k73h6oTkMd4M3wnzhuyyq04VVY=
  private key: (hidden)
  listening port: 51820

peer: D9sP9L1Q+pKtgok10+b+nlojDn6PxCyFMwGHZ/p+i4=
  endpoint: 10.42.1.22:51820
  allowed ips: 10.23.0.1/32
  latest handshake: 10 seconds ago
  transfer: 124 B received, 180 B sent
  persistent keepalive: every 25 seconds

```

```

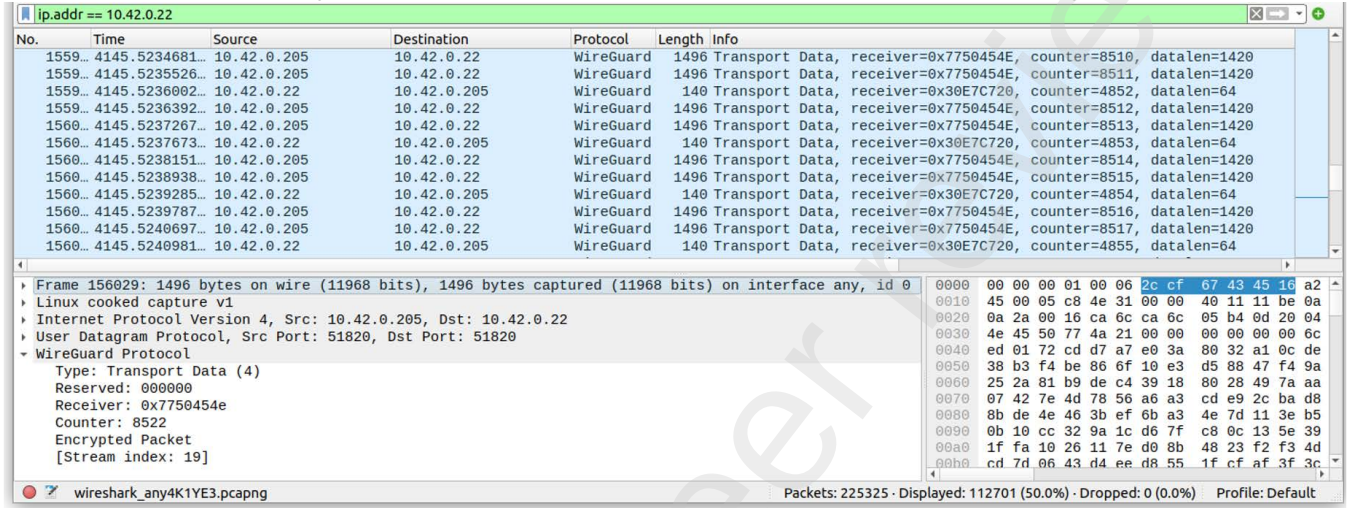
CT@dect-gnbsim$ sudo wg
interface: wg0
  public key: D9sP9L1Q+pKtgok10+b+nlojDn6PxCyFMwGHZ/p+i4=
  private key: (hidden)
  listening port: 51820

peer: YBgIvD3LeiLQC/35k73h6oTkMd4M3wnzhuyyq04VVY=
  endpoint: 10.42.1.205:51820
  allowed ips: 10.23.0.2/32
  latest handshake: 6 seconds ago
  transfer: 180 B received, 272 B sent
  persistent keepalive: every 25 seconds

```

(a) WireGuard configuration at UE1.

(b) WireGuard configuration at UE2.



(c) Wireshark capture of Uni-RAT WireGuard-protected data-plane transmission.

Fig. 17: Uni-RAT WireGuard configuration and data-plane operation: (a) WireGuard configuration at UE1, (b) WireGuard configuration at UE2, and (c) WireGuard-protected data transmission observed in Wireshark.

bottleneck is primarily hardware-limited and is expected to improve with wider channel bandwidths or next-generation Wi-Fi technologies (e.g., Wi-Fi 7/8).

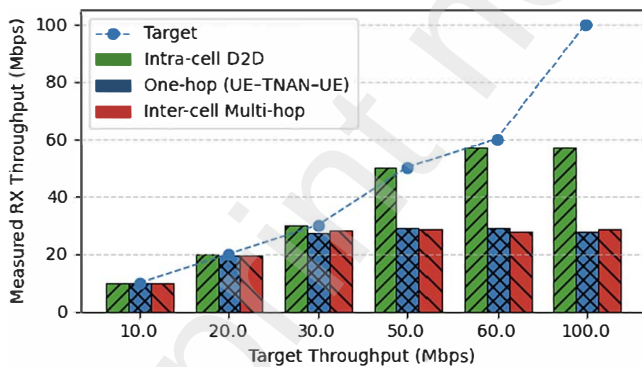


Fig. 18: Iperf3 bandwidth testing on the Uni-RAT system.

Fig. 19 reports the average UDP packet loss observed during the experiments. As shown, intra-cell D2D communication exhibits the lowest packet loss, remaining close to zero. In contrast, the one-hop and inter-cell multi-hop scenarios experience higher packet loss, though it remains within an acceptable range (below 0.5%). This behavior is primarily attributed

to additional relay hops, where packets traverse the TNAN, introducing buffering constraints, processing overhead, and increased exposure to wireless interference. As a result, packet loss increases with the number of forwarding nodes, which is consistent with practical multi-hop wireless communication systems. These results further indicate that the proposed Uni-RAT framework supports not only high throughput but also strong communication reliability.

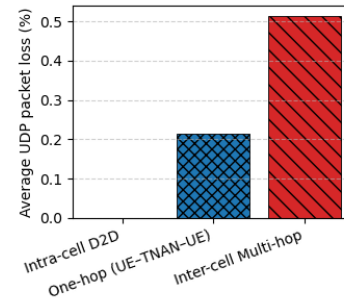


Fig. 19: Packet loss performance of the Uni-RAT system.

Fig. 20 reports the ping RTT and jitter measured during the experiments. As shown in Fig. 20a, intra-cell D2D communication achieves the lowest RTT, while the one-hop

scenario exhibits higher RTT than the inter-cell multi-hop case. This is because, in the one-hop scenario, UE traffic traverses the TNAN access point over CSMA/CA based Wi-Fi, where both UEs share the same channel and experience contention on both uplink and downlink, resulting in increased delay. In contrast, in the inter-cell multi-hop scenario, the two UEs are associated with different access points operating on different Wi-Fi channels, thereby avoiding contention and reducing access delay despite the longer path. Fig. 20b shows a similar trend for jitter. Due to the random nature of channel contention under CSMA/CA, the one-hop scenario experiences higher jitter compared to both D2D and inter-cell multi-hop communication. Although the multi-hop scenario involves a longer transmission path with additional relays, its RTT and jitter remain more stable because contention is distributed across different channels. Overall, these results confirm that the proposed Uni-RAT framework adapts well to diverse deployment scenarios, enabling reliable D2D communication with acceptable latency and jitter for practical deployments.

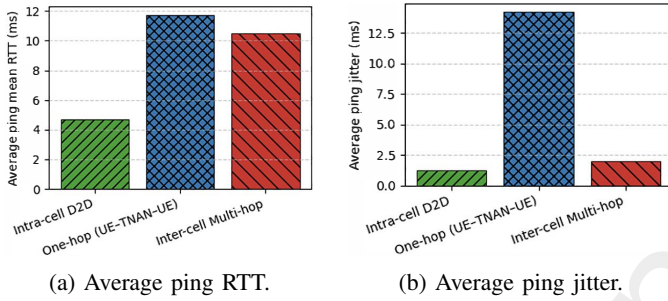


Fig. 20: Uni-RAT ping test with different scenarios.

2) *Uni-RAT Core Network Evaluation:* To further evaluate the Uni-RAT system, the developed Cumucore (CU) network was tested under single UE registration and up to 100 concurrent UE registrations, measuring UE registration latency in terms of the mean and standard error of the mean (SEM). In addition, Open5GC was used as a reference platform to assess the impact of the developed core network applications on UE registration latency.

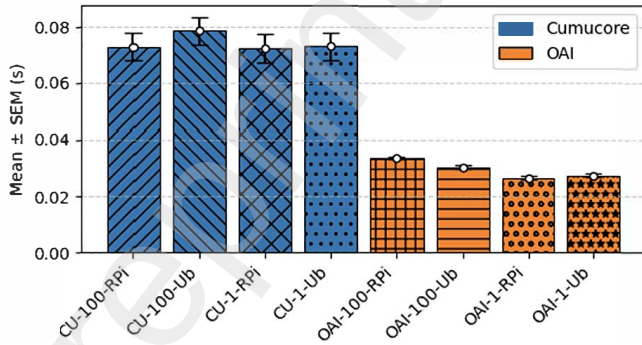


Fig. 21: 100 UEs Authentication on CU and OAI Open5GC.

Fig. 21 summarizes the UE registration latency, reported as the mean and SEM in seconds, based on 100 test repetitions

for each configuration. Both platforms achieve registration latency below 0.1 s, which is acceptable for practical deployments. The Cumucore network exhibits slightly higher latency and a larger SEM than OAI Open5GC, reflecting additional processing overhead from management GUIs, network slicing support, and other core network procedures beyond basic UE registration, whereas OAI Open5GC completes registration at the AMF stage. No significant difference is observed between single-UE and 100-UE scenarios on either platform, as registration requests are effectively handled in a serialized manner due to core network signaling and processing constraints. Furthermore, heterogeneous UEs running on different hardware platforms and operating systems show comparable registration times, as the Uni-RAT UE and TNAN are implemented in C and designed for cross-platform compatibility, demonstrating the practicality and scalability of the proposed Uni-RAT system even under stressed registration conditions.

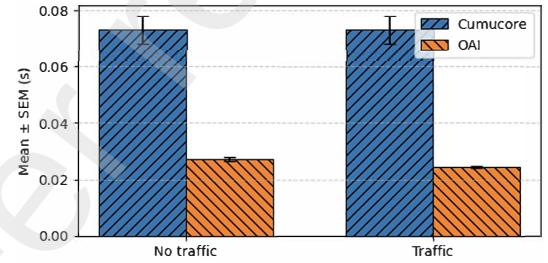


Fig. 22: Background traffic impact on authentication latency.

Fig. 22 illustrates the impact of background traffic on core network authentication time, measured using one UE over 100 authentication trials under 1 Gbps background traffic conditions. The results show that background traffic does not degrade authentication time, indicating that control-plane authentication procedures remain stable under concurrent data-plane load for both Cumucore and OAI Open5GC systems. In terms of variability, the SEM remains comparable for Cumucore, while a slight reduction is observed for OAI, which is likely attributable to system-level effects such as CPU frequency scaling and warm service states under sustained traffic load, rather than a fundamental performance improvement.

TABLE III: WireGuard vs. IPsec-Based VPNs

Metric	WG	IPsec	Gain
Throughput	High	≈50% lower	>2×
RTT	Low	High	~30–50%
CPU load	Low	High	~40–60%
Code size	~4k	>60k	~15×
Handshake	Stateless	IKEv2 + SA	–

D. Comparison Between WireGuard and IPsec

WireGuard and IPsec follow fundamentally different VPN design philosophies with clear performance implications, as summarized in Table III. Experimental results by Mackey et

TABLE IV: Comparison of Non-3GPP RAT Cooperation and Integration Approaches

System	RAT Techniques	Bandwidth	Range	Security
AirDrop [29]	BLE + Wi-Fi Direct	High ($\sim$ Mbps)	Short ($\sim$ 30 m)	Account-based
LoRaBee [30]	LoRa + ZigBee (PHY)	Low ($\sim$ bps)	Long ($\sim$ km)	–
LoFi [34]	Wi-Fi 6 + LoRa (PHY)	Medium ($\sim$ kbps)	Long ($\sim$ km)	–
EWDC [7]	DECT-2020 NR + Wi-Fi	High ($\sim$ Mbps)	Short ($\sim$ 30 m)	Secure control plane
Uni-RAT	DECT-2020 NR + Wi-Fi	High ($\sim$Mbps)	Adaptive	Cellular NAS / 5G-AKA

al. [60] show that WireGuard achieves more than $2\times$ higher throughput, 30–50% lower RTT, and 40–60% lower CPU utilization than OpenVPN, which shares key architectural characteristics with IPsec-based deployments. These gains are primarily enabled by WireGuard’s kernel-space implementation, compact codebase, and native multi-threading, allowing efficient scaling on multi-core systems. In contrast, IPsec relies on IKEv2 negotiation, multiple security associations (SA), and complex policy management, resulting in higher control-plane latency and sensitivity to context switching, particularly in multi-session and multi-RAT scenarios. Moreover, WireGuard’s stateless, noise-based cryptographic design significantly reduces handshake complexity and setup overhead, making it a more efficient alternative to IPsec for scalable, low-latency, and high-throughput data-plane protection in next-generation multi-RAT and 6G systems.

E. Comparison of Uni-RAT with other Systems

Table IV compares Uni-RAT with representative non-3GPP RAT cooperation approaches. Commercial systems such as AirDrop [29], Google Nearby Share, and Intel Unison combine BLE-based discovery with Wi-Fi Direct to achieve Mbps-level throughput over short ranges, which limits scalability and adds latency. Low-power cross-technology solutions, including LoRaBee [30] and LoFi [34], emphasize PHY-layer interoperability, enabling kilometer-scale or long-range communication at bps–kbps rates without integrated security. In contrast, Uni-RAT, evolved from the EWDC architecture [7], integrates DECT-2020 NR and Wi-Fi under a unified 3GPP security framework using NAS signaling and 5G-AKA, supports high-throughput Wi-Fi data delivery, and enables secure cross-cell communication via SMS-over-NAS and one-hop and multi-hop WireGuard, providing a scalable, network-grade foundation for heterogeneous multi-RAT integration.

VI. CONCLUSION AND FUTURE WORK

This paper presented Uni-RAT, a unified Multi-RATs framework that addresses the fragmentation and security limitations of current heterogeneous access architectures by consolidating control, authentication, and security across non-3GPP RATs. By designating DECT-2020 NR as a Master RAT, Uni-RAT preserves a secure 3GPP connection with the 5G core, performs 5G-AKA authentication, and distributes the NAS security context to Secondary-RATs such as Wi-Fi, which provide high-throughput data-plane connectivity. To reduce the overhead and complexity of conventional IPsec-based

interworking, Uni-RAT introduces a lightweight WireGuard-based Layer-3 data plane, enabling efficient secure communication between the core and devices as well as direct D2D connectivity. A prototype implementation using Nordic nRF9161 devices, Raspberry Pi platforms, and a Cumucore 5GC demonstrates stable operation and scalability with up to 100 concurrent RAT devices. These results indicate that Uni-RAT offers a practical, scalable, and security-centric foundation for integrating non-3GPP RATs into future 6G heterogeneous access networks.

Future work will extend Uni-RAT to support Wi-Fi 7 and DECT-2020 NR low-energy operation, enabling higher throughput and improved energy efficiency in heterogeneous access deployments. We will also investigate large-scale WG-AF deployments in the core network, including scalability analysis and performance evaluation under massive device populations, with a focus on efficient WireGuard-based data plane management, optimized key distribution, and reduced processing overhead. In addition, Uni-RAT will be expanded to integrate other RATs, such as BLE and LoRa, as either Master or Secondary-RATs, enabling flexible Multi-RATs configurations while further strengthening security through unified 3GPP authentication and control mechanisms.

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APPENDIX A MULTI-HOP WIREGUARD CONFIGURATION

```
[Interface]
# WireGuard interface on Uni-RAT TNAN1
PrivateKey = <private-key-TNAN1>
Address    = 10.0.0.1/32
ListenPort = 51820

[Peer]
# Relay switch (hub)
PublicKey  = <public-key-RelaySwitch>
Endpoint   = 192.168.88.1:51820
AllowedIPs = 0.0.0.0/0, ::/0
PersistentKeepalive = 15

[Peer]
# Uni-RAT UE1 (STA1)
PublicKey  = <public-key-UE1>
AllowedIPs = 10.0.0.5/32
```

Listing 1: TNAN1 (AP1) WireGuard Configuration

```
[Interface]
# WireGuard interface on Uni-RAT UE1
PrivateKey = <private-key-UE1>
Address    = 10.0.0.5/32

[Peer]
# Uni-RAT TNAN1 (AP1)
PublicKey  = <public-key-TNAN1>
Endpoint   = 10.42.0.1:51820
AllowedIPs = 10.0.0.0/24, 10.42.1.0/24
PersistentKeepalive = 15
```

Listing 2: UE1 (STA1) WireGuard Configuration

```
[Interface]
# WireGuard interface on Uni-RAT TNAN2
PrivateKey = <private-key-TNAN2>
Address    = 10.0.0.3/32
ListenPort = 51820

[Peer]
# Relay switch (hub)
PublicKey  = <public-key-RelaySwitch>
Endpoint   = 192.168.88.1:51820
AllowedIPs = 0.0.0.0/0, ::/0
PersistentKeepalive = 15

[Peer]
# Uni-RAT UE2 (STA2)
PublicKey  = <public-key-UE2>
AllowedIPs = 10.0.0.4/32
```

Listing 3: TNAN2 (AP2) WireGuard Configuration

```
[Interface]
# WireGuard interface on Uni-RAT UE2
PrivateKey = <private-key-UE2>
Address    = 10.0.0.4/32

[Peer]
# Uni-RAT TNAN2 (AP2)
PublicKey  = <public-key-TNAN2>
Endpoint   = 10.42.1.1:51820
AllowedIPs = 10.0.0.0/24, 10.42.0.0/24
PersistentKeepalive = 15
```

Listing 4: UE2 (STA2) WireGuard Configuration

```
[Interface]
# WireGuard interface on the switch
PrivateKey = <private-key-RelaySwitch>
Address    = 10.0.0.2/32
ListenPort = 51820

[Peer]
# Uni-RAT TNAN1 (AP1)
PublicKey  = <public-key-TNAN1>
AllowedIPs = 10.0.0.1/32, 10.42.0.0/24, 10.0.0.5/32
PersistentKeepalive = 15

[Peer]
# Uni-RAT TNAN2 (AP2)
PublicKey  = <public-key-TNAN2>
AllowedIPs = 10.0.0.3/32, 10.42.1.0/24, 10.0.0.4/32
PersistentKeepalive = 15
```

Listing 5: WireGuard Configuration on Switch

```
#!/usr/sbin/nft -f

flush ruleset

table ip nat {
    chain prerouting {
        type nat hook prerouting priority -100;
    }

    chain postrouting {
        type nat hook postrouting priority 100;
        # Masquerade all packets leaving wlan0 (
        # hotspot interface)
        oifname "wlan0" masquerade
    }
}

table ip filter {
    chain input {
        type filter hook input priority 0;
        accept
    }

    chain forward {
        type filter hook forward priority 0;
        accept
    }

    chain output {
        type filter hook output priority 0;
        accept
    }
}
```

Listing 6: nftables configuration for NAT and forwarding on Uni-RAT relay node

```
sudo systemctl enable nftables
sudo systemctl restart nftables
```

Listing 7: Enable and Restart nftables Service